Example 6.15

The following data is available for the voltage series feedback amplifier of Fig. 6.32.

$$R_B = 50\text{ k}\Omega \quad R_E = 1\text{ k}\Omega \quad R_C = 1\text{ k}\Omega$$
$$h_{ie} = 2\text{ k}\Omega \quad h_{fe} = 150$$

Calculate

(a) A_V and A_{Vf}
(b) R_i and R_{if}
(c) R_o and R_{of}

Solution

(a)
$$A_V = \frac{h_{fe}R_E}{h_{ie}} = \frac{(150)\ (1\text{k}\Omega)}{2\text{k}\Omega} = 75$$
$$1 + \beta A_V = 1 + 75 = 76 \quad [\text{since } \beta = 1]$$
$$A_{Vf} = \frac{A_V}{1 + \beta A_V} = \frac{75}{76} = 0.986$$

(b)
$$R_i = R_B \| h_{ie}$$
$$= 50\text{ k}\Omega \| 2\text{ k}\Omega = 1.923\text{ k}\Omega$$
$$R_{if} = R_i\,[1 + \beta A_V]$$
$$= (1.923\text{ k}\Omega)\,(76) = 146.148\text{ k}\Omega$$

Alternatively

$$R_{if} \approx h_{ie} + h_{fe}\,R_E$$
$$= 2\text{ k}\Omega + (150)\ (1\text{ k}\Omega)$$
$$= 152\text{ k}\Omega$$

(c)
$$R_o = R_E = 1\text{ k}\Omega$$
$$R_{of} = \frac{R_o}{1 + \beta A_V}$$
$$= \frac{1\text{k}\Omega}{76} = 13.15\ \Omega$$

Alternatively

$$R_{of} = \frac{h_{ie}}{h_{fe}} = \frac{2\text{ k}\Omega}{150} = 13.33\ \Omega$$

16.12.3 Current Series Feedback Amplifier

Figure 6.39 shows the circuit of current series feedback amplifier. Note that the circuit is a CE amplifier with unbypassed R_E.

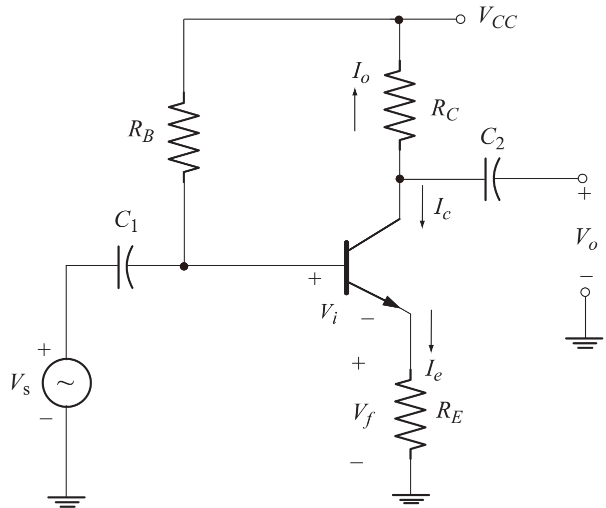


Fig. 6.39 Current series feedback amplifier

The feedback voltage is the voltage across R_E

$$V_f = I_e R_E \tag{6.160}$$

$$I_e \approx I_c$$

But $$I_c = -I_o$$

$\therefore$ $$I_e = -I_o \tag{6.161}$$

Using this relation in Equation (6.160) we have

$$V_f = -R_E I_o \tag{6.162}$$

Equation (6.162) is of the form

$$V_f = \beta I_o \tag{6.163}$$

We find that $$\beta = \frac{V_f}{I_o} = -R_E \tag{6.164}$$

Note that the feedback voltage is proportional to the output current. Hence the sampled signal is the output current.

Applying KVL to the input circuit we have

$$V_s - V_i - V_f = 0$$

or $$V_i = V_s - V_f \tag{6.165}$$

Note that V_f is mixed in series with V_S, also V_f subtracts from V_S which implies that feedback is negative.

The sampled signal is the output current and the feedback voltage derived from the output current is mixed in series with the source voltage. Hence the topology is current series feedback. Also the basic amplifier is a transconductance amplifier.

The ac equivalent circuit is obtained by reducing V_{CC} to zero and short circuiting the capacitors C_1 and C_2 as shown in Fig. 6.40.

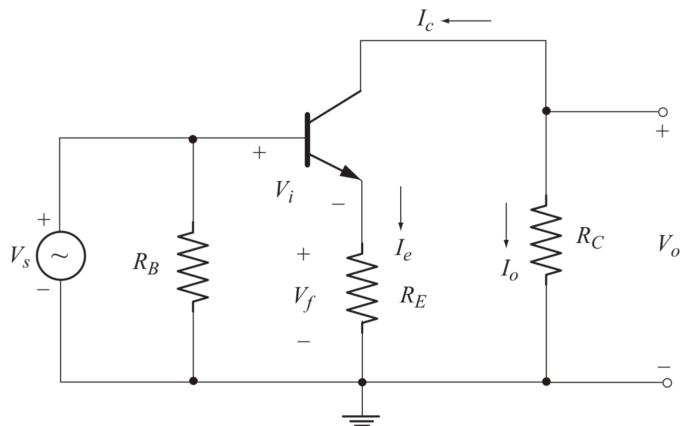


Fig. 6.40 *AC equivalent circuit of current series feedback amplifier*

Basic Amplifier Circuit without Feedback

The input circuit is obtained by reducing the sampling to zero. Since the output current is sampled, it can be reduced to zero by open circuiting the collector in the circuit of Fig. 6.40. This is shown in Fig. 6.41.

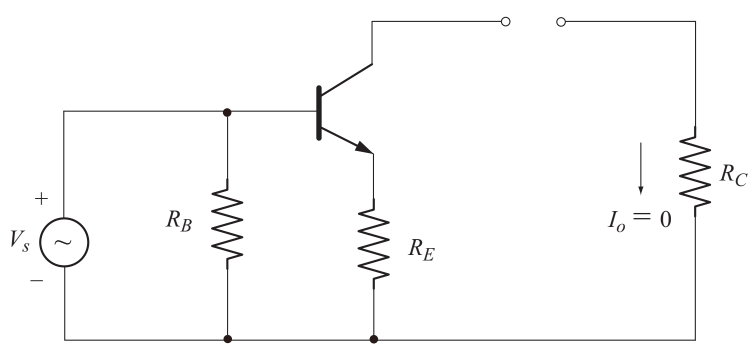


Fig. 6.41 Input circuit obtained by open circuiting the collector

The output circuit is obtained by reducing the mixing to zero. Since the mixing is series, the input loop has to be opened in the circuit of Fig. 6.40. This is shown in Fig. 6.42.

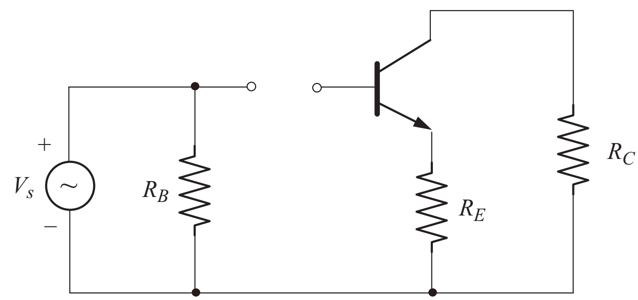


Fig. 6.42 Output circuit obtained by opening the input loop

The basic amplifier circuit without feedback is obtained by combining the features of input and output circuits as shown in Fig. 6.43.

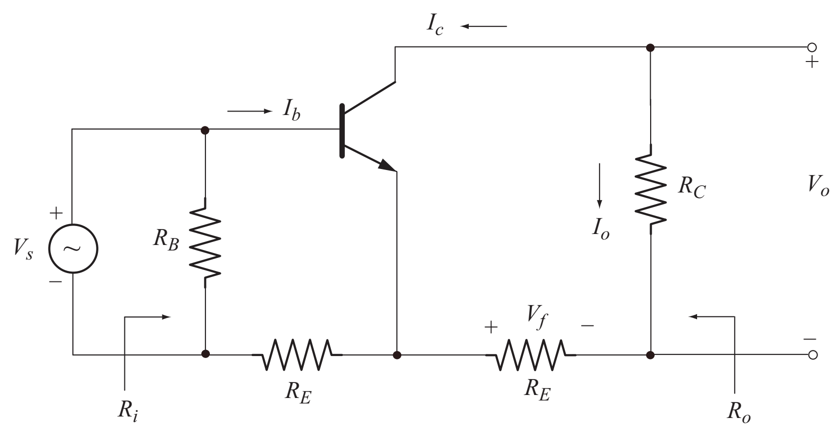


Fig. 6.43 Basic amplifier circuit without feedback

Let us replace the transistor by its small signal approximate hybrid equivalent circuit as shown in Fig. 6.44.

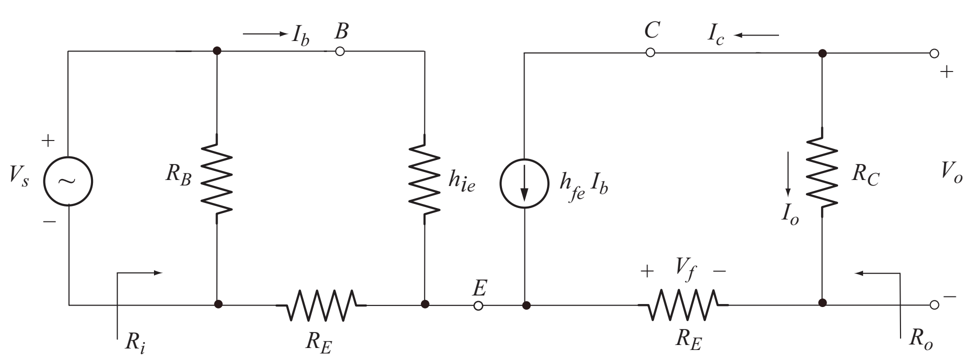


Fig. 6.44 AC equivalent circuit using hybrid model

Transfer Gain with Feedback

Since the basic amplifier is transconductance amplifier the transfer gain is the transcondcutance Transfer gain without feedback is given by

$$A = G_M = \frac{I_o}{V_i} \tag{6.166}$$

From Equation (6.165)

$$V_i = V_s - V_f$$

In the absence of feedback, $V_f = 0$

$$\therefore \qquad V_i = V_s$$

Now from Equation 6.166 we have

$$A = \frac{I_o}{V_s} \tag{6.167}$$

$$\text{But} \qquad I_o = -I_c$$

$$\text{and} \qquad I_c = h_{fe} I_b$$

$$\therefore \qquad I_o = -h_{fe} I_b \tag{6.168}$$

Applying KVL to the input circuit of Fig. 6.44 we have

$$V_s = I_b [h_{ie} + R_E] \tag{6.169}$$

Using Equations (6.168) and (6.169) in Equation (6.167) we have

$$A = \frac{-h_{fe} I_b}{I_b [h_{ie} + R_E]}$$

$$\text{Now} \qquad A = G_M = \frac{-h_{fe}}{h_{ie} + R_E} \tag{6.170}$$

Transfer gain with feedback is

$$A_f = G_{Mf} = \frac{I_o}{V_s} = \frac{A}{1 + \beta A} \tag{6.171}$$

$$1 + \beta A = 1 + [-R_E] \left[\frac{-h_{fe}}{h_{ie} + R_E} \right]$$

$$= \frac{h_{ie} + (1 + h_{fe}) R_E}{h_{ie} + R_E} \tag{6.172}$$

$$\text{Now} \qquad A_f = \frac{\left[\dfrac{-h_{fe}}{h_{ie} + R_E} \right]}{\left[\dfrac{h_{ie} + (1 + h_{fe}) R_E}{h_{ie} + R_E} \right]}$$

$$A_f = G_{Mf} = \frac{-h_{fe}}{h_{ie} + (1 + h_{fe})R_E} \tag{6.173}$$

Since $\quad h_{fe} \gg 1,\ (1 + h_{fe}) \approx h_{fe}$

Now $$A_f = G_{Mf} \approx \frac{-h_{fe}}{h_{ie} + h_{fe} R_E} \tag{6.174}$$

Input Resistance with Feedback

From the circuit of Fig. 6.44, the input resistance without feedback is

$$R_i = R_B \parallel [h_{ie} + R_E]$$

Usually $\quad R_B \gg h_{ie} + R_E$

$$\therefore \quad R_i \approx h_{ie} + R_E \tag{6.175}$$

Series mixing increases input resistance. Therefore input resistance with feedback is

$$R_{if} = R_i [1 + \beta A]$$

$$= [h_{ie} + R_E] \left[\frac{h_{ie} + (1 + h_{fe}) R_E}{h_{ie} + R_E} \right]$$

$$R_{if} = h_{ie} + (1 + h_{fe}) R_E \tag{6.176}$$

Since $\quad 1 + h_{fe} \approx h_{fe}$

$$R_{if} \approx h_{ie} + h_{fe} R_E \tag{6.177}$$

Output Resistance with Feedback

To find the output resistance without feedback, let us reduce the current source to zero in the output circuit of Fig. 6.44. The resulting circuit is shown in Fig. 6.45.

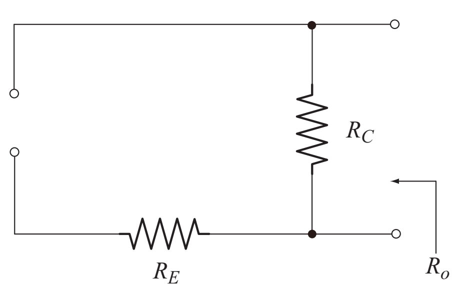


Fig. 6.45 Circuit to find output resistance

From the circuit of Fig. 6.45, the output resistance without feedback is

$$R_o = R_C \tag{6.178}$$

Current sampling increases output resistance. Therefore output resistance with feedback is

$$R_{of} = R_o[1 + \beta A]$$

$$R_{of} = R_C\left[\frac{h_{ie} + (1 + h_{fe})R_E}{h_{ie} + R_E}\right]$$

$$= R_C\left[\frac{(h_{ie} + R_E) + h_{fe}R_E}{h_{ie} + R_E}\right]$$

or

$$R_{of} = R_C\left[1 + \frac{h_{fe}R_E}{h_{ie} + R_E}\right] \tag{6.179}$$

Voltage Gain with Feedback

Voltage gain without feedback is

$$A_V = \frac{V_o}{V_i} = \frac{V_o}{V_s}$$

But

$$V_o = I_o R_C$$

$\therefore$

$$A_V = \left[\frac{I_o}{V_s}\right] R_C \tag{6.180}$$

But

$$A = \frac{I_o}{V_s}$$

$\therefore$

$$A_V = A\, R_C = G_M\, R_C \tag{6.181}$$

Substituting for A from Equation (6.170) we have

$$A_V = \frac{-h_{fe}R_C}{h_{ie} + R_E} \tag{6.182}$$

Using Equation (6.181), the voltage gain with feedback can be written as

$$A_{Vf} = A_f\, R_C = G_{Mf}\, R_C \tag{6.183}$$

Substituting for A_f from Equation (6.174) we have

$$A_{Vf} = \frac{-h_{fe}R_C}{h_{ie} + h_{fe}R_E} \tag{6.184}$$

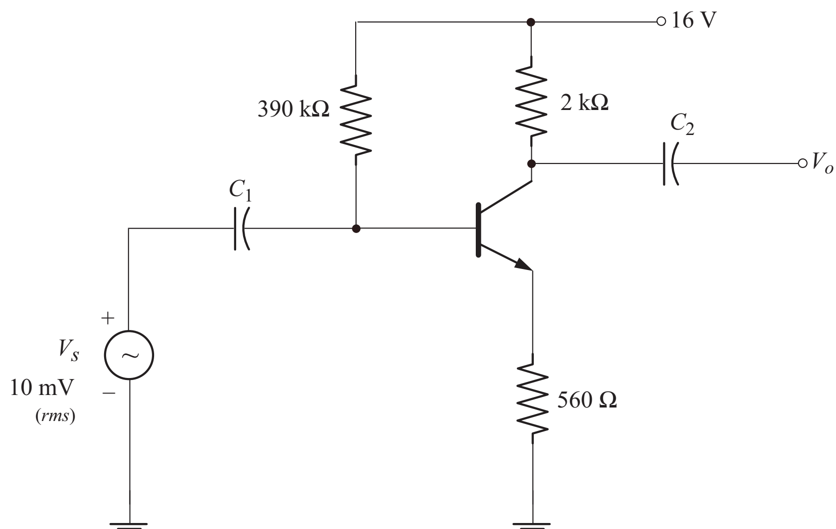
Example 6.16

For the current series feedback amplifier shown below, calculate the following:

(a) Desentisivity factor
(b) Transfer gain with feedback
(c) Voltage gain with and without feedback

(d) Input resistance with and without feedback
(e) Output resistance with and without feedback
(f) Output current with and without feedback
(g) Output voltage with and without feedback

For the transistor used, $h_{fe} = 200$ and $h_{ie} = 2\ \text{k}\Omega$.



(a) Desentisivity factor is

$$D = 1 + \beta A$$

$$\beta = -R_E = -560\ \Omega$$

$$A = G_M = \frac{-h_{fe}}{h_{ie} + R_E}$$

$$= \frac{-200}{2000\ \Omega + 560\ \Omega} = -\ 0.078125\ \mho$$

$$D = 1 + [-560\ \Omega]\ [-\ 0.078125\ \mho] = 44.75$$

(b) Transfer gain with feedback is

$$A_f = G_{Mf} = \frac{A}{D} = \frac{-\ 0.078125\ \mho}{44.75} = -1.745\ \text{m}\mho$$

(c) Voltage gain without feedback is

$$A_V = A\,R_C = [-\ 0.078125\ \mho\ ]\ [2\ \text{k}\Omega] = -\ 156.25$$

Voltage gain with feedback is

$$A_{Vf} = A_f\,R_C = (-\ 1.745\ \text{m}\mho)\ (2\ \text{k}\Omega) = -\ 3.49$$

(d) Input resistance without feedback is

$$R_i = h_{ie} + R_E = 2\text{ k}\Omega + 560\ \Omega = 2.56\text{ k}\Omega$$

Input resistance with feedback is

$$R_{if} = h_{ie} + h_{fe} R_E = 2\text{ k}\Omega + (200)\ (560\ \Omega) = 114\text{ k}\Omega$$

Alternatively

$$R_{if} = R_i\ [1 + \beta A] = (2.56\text{ k}\Omega)\ (44.75) = 114.56\text{ k}\Omega$$

(e) Output resistance with out feedback is

$$R_o = R_C = 2\text{ k}\Omega$$

Output resistance with feedback is

$$R_{of} = R_C \left[1 + \frac{h_{fe} R_E}{h_{ie} + R_E}\right]$$

$$= (2\text{ k}\Omega) \left[1 + \frac{(200)\ (560\Omega)}{2\text{k}\Omega + 560\Omega}\right] = 89.5\text{ k}\Omega$$

Alternatively

$$R_{of} = R_o\ [1 + \beta A] = (2\text{ k}\Omega)\ (44.75) = 89.5\text{ k}\Omega$$

(f) Output current without feedback is

$$I_o = A \cdot V_S$$
$$= (-0.078125\ \Omega)\ (10\text{ mV})$$
$$= -\ 0.78125\text{ mA (rms)}$$

Output current with feedback is

$$I_o = A_f V_s$$
$$= (-1.745\text{ m℧})\ (10\text{ mV}) = -17.45\ \mu\text{A (rms)}$$

(g) Output voltage with out feedback is

$$V_o = A_V V_s$$
$$= (-156.25)\ (10\text{ mV}) = -1.5625\text{ V}$$

Output voltage with feedback is

$$V_o = A_{Vf} V_s$$
$$= (-3.49)\ (10\text{ mV})$$
$$= -34.9\text{ mV (rms).}$$

Note: If h_{ie} is not given, we can find the same using the relation, $h_{ie} = \beta\, r_e = h_{fe} r_e$

where $$r_e = \frac{26\text{ mV}}{I_E}$$

6.12.4 Voltage-shunt Feedback Amplifier

Figure 6.46 shows the voltage shunt feedback amplifier using an FET. Feed back is given from drain to gate through the feedback resister R_F.

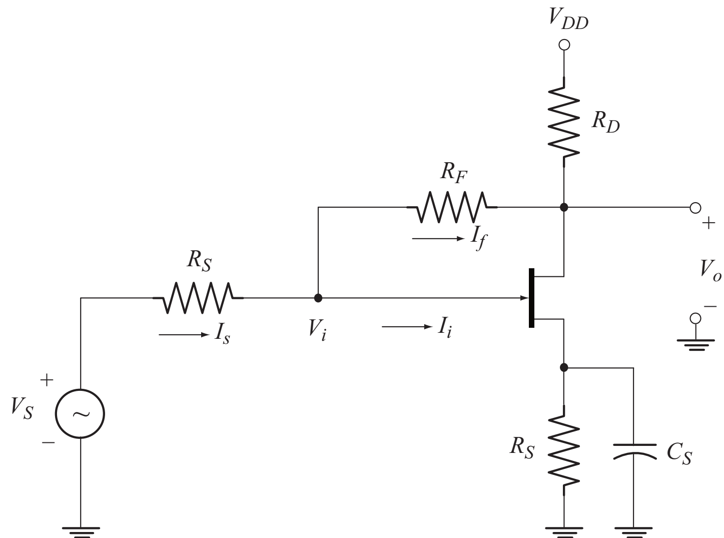


Fig. 6.46 Voltage-shunt feedback amplifier

The feedback current is given by

$$I_f = \frac{V_i - V_o}{R_F} \tag{6.185}$$

Due to voltage amplification,

$$|V_o| \gg |V_i|$$

$$\therefore \qquad I_f \approx \frac{-V_o}{R_F} \tag{6.186}$$

The above equation is of the form

$$I_f = \beta V_o \tag{6.187}$$

Comparing the two equations we find that

$$\beta = -\frac{1}{R_F} \tag{6.188}$$

Note that the feedback current is derived from the output voltage. Hence the sampled signal is the output voltage. Also feedback network converts output voltage into feedback current.

Applying KCL at the input node we have

$$I_S = I_i + I_f$$

$$\text{or} \qquad I_i = I_S - I_f \tag{6.189}$$

Note that the source current is mixed in shunt with the feedback current. Also the feedback current subtracts from the source current. Hence the feedback is negative.

Since the output voltage is sampled and the feedback current derived from the output voltage is mixed in shunt with source current, the topology is voltage-shunt feedback.

AC Equivalent Circuit of Feedback Amplifier

The ac equivalent circuit of feedback amplifier is obtained by reducing V_{DD} to zero and treating the capacitor C_S as short circuit. The resulting circuit is shown in Fig. 6.47. The voltage source at the input is converted into its equivalent current source.

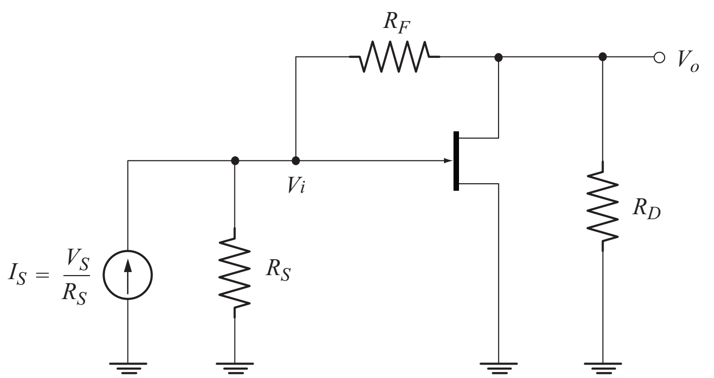


Fig. 6.47 Ac equivalent circuit of voltage-shunt feedback amplifier

Basic Amplifier Circuit without Feedback

The input circuit is obtained by reducing sampling to zero. Since output voltage is sampled, the sampling can be reduced to zero by reducing V_o to zero. This is done by connecting drain terminal to ground. Now as seen from the input side, R_F appears between gate and ground.

The output circuit is obtained by reducing mixing to zero. Since the mixing is shunt, it can be reduced to zero by reducing V_i to zero or connecting gate terminal to ground. Now as seen from the output side R_F appears between drain and ground.

These features are combined to write the basic amplifier circuit with out *feedback* as shown in Fig. 6.48

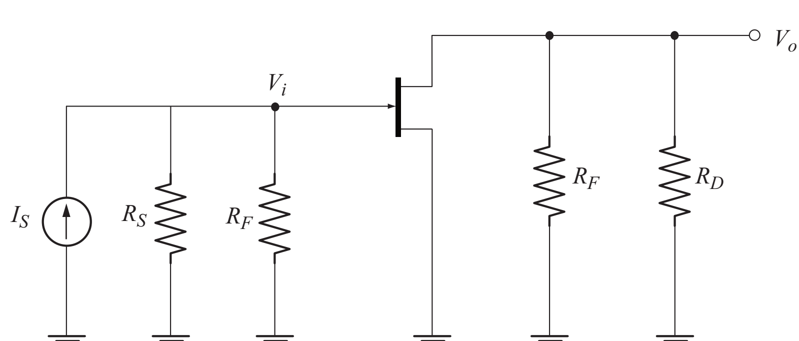


Fig. 6.48 Basic amplifier circuit with out feedback

Let us replace the FET by its small signal model. The resulting circuit is shown in Fig. 6.49.

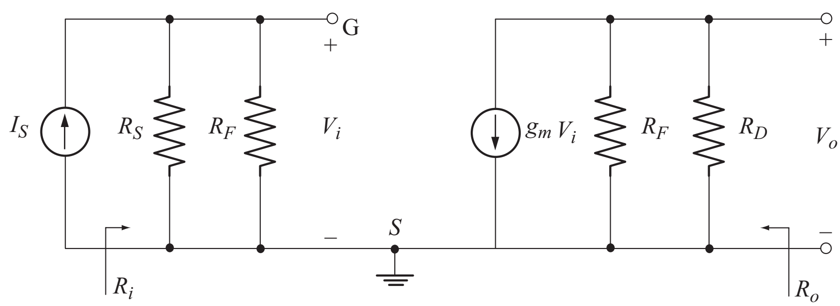


Fig. 6.49 Basic amplifier circuit using small signal model of FET

Transfer Gain with Feedback

Transfer gain with out feedback is given by

$$A = R_M = \frac{V_o}{I_s} \tag{6.190}$$

From the output circuit of Fig. 6.49, we have

$$V_o = -g_m V_i [R_F \,||\, R_D] \tag{6.191}$$

usually $R_F \gg R_D$. As a result

$$R_F \,||\, R_D \approx R_D$$

$$\therefore \quad V_o \approx -g_m V_i R_D \tag{6.192}$$

$$\text{Also} \quad V_i = I_S \, [R_S \,||\, R_F]$$

Taking $R_S \,||\, R_F \approx R_S$, we have

$$V_i \approx I_S R_S = V_S \tag{6.193}$$

using Equation (6.193) in Equation (6.192) we have

$$V_o = -g_m [I_S R_S] R_D$$

$$\text{Now} \quad A = \frac{V_o}{I_s} = -g_m R_S R_D \tag{6.194}$$

Transfer gain with feedback is given by

$$A_f = R_{Mf}$$

$$= \frac{V_o}{I_s} = \frac{A}{1+\beta A}$$

$$A_f = \frac{-g_m R_D R_S}{1+\left[\frac{-1}{R_F}\right][-g_m R_D R_S]}$$

$$A_f = \frac{-g_m R_D R_S R_F}{R_F + g_m R_D R_S} \tag{6.195}$$

Voltage Gain with Feedback

From Equation (6.192)

$$V_o = -g_m V_i R_D \tag{6.196}$$

Using Equation (6.193) in Equation (6.196) we have

$$V_o = -g_m V_s R_D$$

Now voltage gain without feedback is

$$A_V = \frac{V_o}{V_s} = -g_m R_D \tag{6.197}$$

Voltage gain with feedback is

$$A_{Vf} = \frac{V_o}{V_s}$$

$$= \left[\frac{V_o}{I_s}\right]\left[\frac{I_s}{V_s}\right]$$

$$= [A_f]\left[\frac{1}{R_s}\right] \qquad [\because V_S = I_S R_S]$$

Substituting for A_f from Equation (6.195) we have

$$A_{Vf} = \left[\frac{-g_m R_D R_S R_F}{R_F + g_m R_D R_S}\right]\left[\frac{1}{R_S}\right]$$

$$A_{Vf} = \frac{-g_m R_D R_F}{R_F + g_m R_D R_S} \tag{6.198}$$

Input Impedance with Feedback

From the circuit of Fig. 6.49, input impedance without feedback is

$$R_i = R_S \,||\, R_F \tag{6.199}$$

Shunt mixing reduces input impedance. Hence input impedance with feedback is

$$R_{if} = \frac{R_i}{1+\beta A} \tag{6.200}$$

Output Impedance with Feedback

From the circuit of Fig. 6.49, output impedance without feedback is

$$R_o = R_F \parallel R_D \tag{6.201}$$

Voltage sampling reduces output impedance. Hence output impedance with feedback is

$$R_{of} = \frac{R_o}{1+\beta A} \tag{6.202}$$

Example 6.17

The following data are available for the voltage shunt feedback amplifier of Fig. 6.46.

$g_m = 5$ m℧ $R_D = 5.1$ kΩ $R_S = 1$ kΩ $R_F = 10$ kΩ

Calculate the following:

(a) Transfer gain with and without feedback
(b) Voltage gain with and without feedback
(c) Input impedance with and without feedback
(d) Output impedance with and without feedback

Solution

(a) Transfer gain without feedback is

$$\begin{aligned} A &= -g_m R_D R_S \\ &= -[5\text{ m℧}]\,[5.1\text{ k}\Omega]\,[1\text{ k}\Omega] = -25.5\text{ k}\Omega \end{aligned}$$

Transfer gain with feedback is

$$A_f = \frac{A}{1+\beta A}$$

$$1+\beta A = 1 + \left[\frac{-1}{10\text{k}\Omega}\right][-25.5\text{ k}\Omega] = 3.55$$

$$A_f = \frac{-25.5\text{k}\Omega}{3.55} = -7.18\text{ k}\Omega$$

(b) Voltage gain with out feedback is

$$A_V = -g_m R_D = -[5\text{ m℧}]\,[5.1\text{ k}\Omega] = -25.5$$

Voltage gain with feedback is

$$A_{Vf} = \frac{A_f}{R_S} = \frac{-7.18\text{k}\Omega}{1\text{k}\Omega} = -7.18$$

(c) Input impedance with out feedback is

$$R_i = R_S \parallel R_F = 1\text{ k}\Omega \parallel 10\text{ k}\Omega = 0.909\text{ k}\Omega$$

Input impedance with feedback is

$$R_{if} = \frac{R_i}{1+\beta A} = \frac{0.909\,\text{k}\Omega}{3.55} = 256\ \Omega$$

(d) output impedance with out feedback is

$$R_o = R_D \,\|\, R_F = 5.1\ \text{k}\Omega \,\|\, 10\ \text{k}\Omega = 3.37\ \text{k}\Omega$$

output impedance with feedback is

$$R_{of} = \frac{R_o}{1+\beta A} = \frac{3.37\text{k}\Omega}{3.55} = 0.949\ \text{k}\Omega$$

Exercise Problems

6.1 A feedback amplifier has a gain $A = -1000$. Calculate the gain with feedback if 10% negative feedback is given.

6.2 An amplifier has a gain of – 200 and gain variation of 15%. Calculate the gain variation if 20% negative feedback is given.

6.3 A voltage series negative feedback amplifier has the following data.

$A = -1000$ $R_i = 2$ kΩ $R_o = 25$ kΩ.

If 20% negative feedback is given, calculate the values of A_f, R_{if} and R_{of}.

6.4 An FET voltage series feedback amplifier has the following data.

$R_1 = 500$ kΩ $R_2 = 1$ kΩ $R_o = 30$ kΩ $R_D = 10$ kΩ and $g_m = 4000$ μS.

Calculate A_V and A_{Vf}.

6.5 A transistor current series feedback amplifier has the following data.

$R_B = 50$ kΩ $R_C = 2.7$ kΩ $R_E = 600$ Ω $V_{CC} = 15$ V $h_{fe} = 200$ $h_{ie} = 1000$ Ω.

Calculate A_V and A_{Vf}.

6.6 An FET Voltage shunt feedback amplifier has the following data.

$R_S = 1.2$ kΩ $R_F = 15$ kΩ $R_D = 10$ kΩ $g_m = 10$ mS.

Calculate A_V and A_{Vf}.

Chapter 7

POWER AMPLIFIERS

A power amplifier in a stereo, radio or television system is intended to deliver a large voltage and current in to a low impedance load such as a loud speaker. In a stereo system the voltage level of a small signal from a radio tuner, tape player or compact disc is first amplified by the input stage of power amplifier which is called the driver. The output voltage of the driver then drives the output stage which is a current amplifier. Due to large voltage and current swings in a power amplifier, the non-linearity of the amplifying device is an importnat consideration. Besides this, conversion efficiency is also of prime concern since the power amplifier converts the dc power of the supply voltage to the ac power delivered to the load. This chapter discusses the various classes of operation, power dissipation and non linear distortion resulting from large signal operation.

◆ 7.1 CLASSIFICATION OF POWER AMPLIFIERS

Power amplifiers are classified based on the location of the quiescent operating point or *Q*-point on the dc load line as follows:

1. Class A power amplifiers.
2. Class B power amplifiers.
3. Class AB power amplifiers.
4. Class C power amplifiers.

A class D amplifier is another class of power amplifier, which is designed to operate with digital or pulse type signals.

7.1.1 Class A Power Amplifier

In class A power amplifier, the *Q*-point is located at the centre of the load line as shown in Fig. 7.1, so that the output signal varies over the full cycle of the input signal.

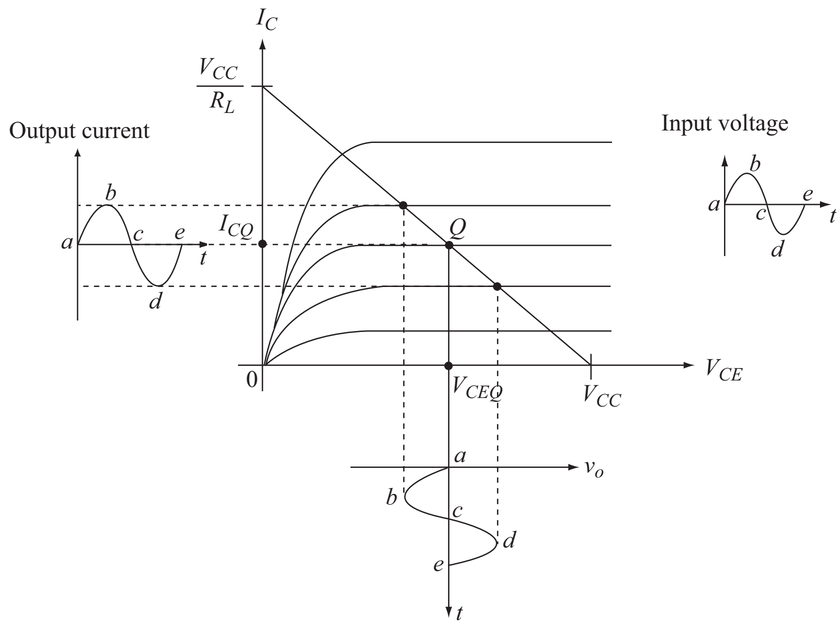


Fig. 7.1 Input-output waveforms for a class A power amplifier

7.1.2 Class B Power Amplifier

In class B power amplifier, the Q-point is located at cut-off as shown in Fig. 7.2, so that the output signal varies over one half cycle of the input signal.

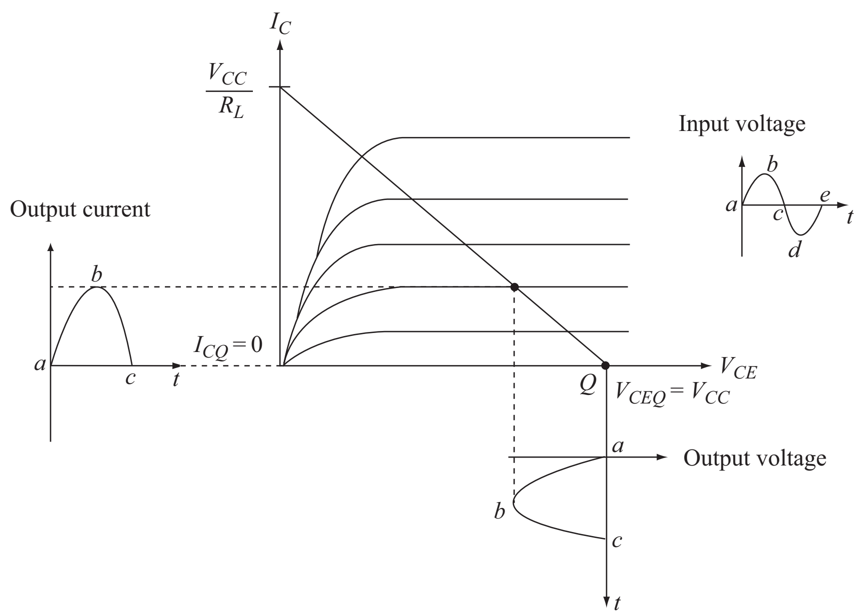


Fig. 7.2 Input-output waveforms for a class B power amplifier

Observe that a single class B amplifier provides only one half-cycle at the output. Thus, a second class B amplifier is used to obtain the amplified output of the other half-cycle. These two amplifiers would then be combined to obtain an amplified output of the full-cycle of the input signal. Such a combination is called a *push-pull configuration*.

7.1.3 Class AB Power Amplifier

We saw in Section 7.1.2 that the *Q*-point of a class B amplifier is at $V_{CE} = V_{CC}$ and $I_C = 0$ mA. This implies that the transistor is at zero bias. The base-emitter junction of the transistor will be forward-biased and the transistor brought into the active region only when the input voltage exceeds the cut-in voltage, V_γ of the base-emitter junction. During the periods when $v_i < V_\gamma$ the output current will be zero as shown in Fig. 7.3.

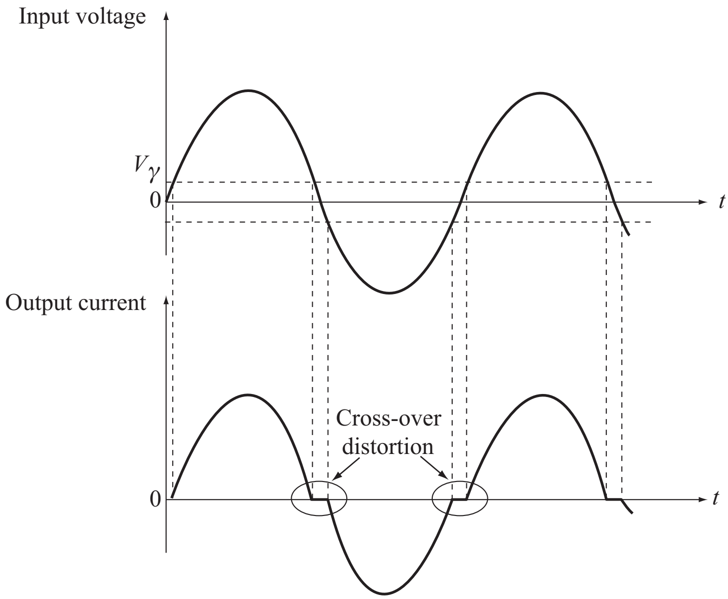


Fig. 7.3 *Cross-over distortion in class B push-pull amplifier*

The output waveform is not an exact replica of the input waveform. i.e., the output signal is distorted. Observe that the distortion occurs at every zero-crossing of the input signal. Hence, the distortion is called cross-over distortion.

This can be overcome by locating the operating point slightly above cut-off. Since, the *Q*-point is located slightly above cut-off as in class B amplifier but much below the centre of the dc load line as in class A amplifier, these amplifiers are referred to as class AB Power Amplifiers.

7.1.4 Class C Power Amplifier

In class C power amplifier, the transistor is biased below cut-off. Let the extent of reverse-bias be V_{BB}. The transistor operates in the active region and a current flows only for $v_i > V_{BB} + V_\gamma$, where V_γ is the cut-in voltage of the base-emitter junction. The current would then be pulses of short duration as shown in Fig. 7.4.

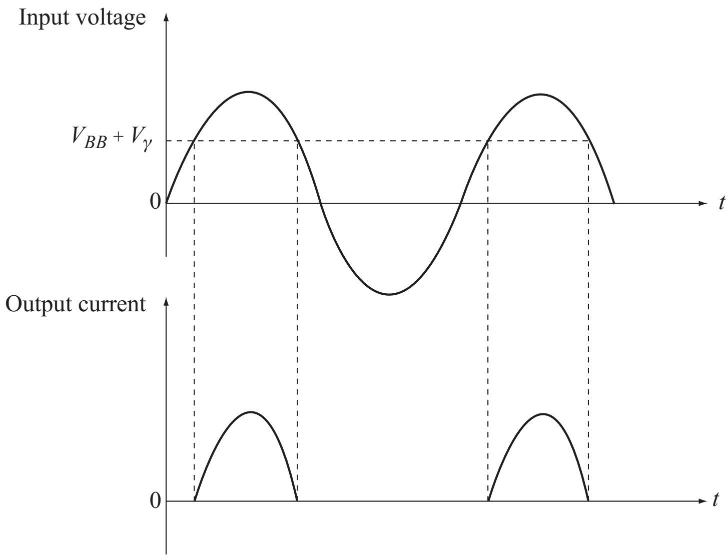


Fig. 7.4 Current pulses in class C power amplifier

Observe that the output current flows for less than one half-cycle of the input signal. The full-cycle of the input signal is obtained at the output by the use of a tuned circuit at the collector as shown in Fig. 7.5.

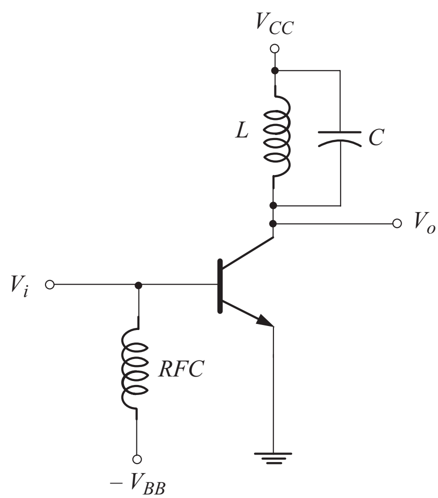


Fig. 7.5 Tuned class C power amplifier

A periodic sinusoidal input signal will produce a periodic non-sinusoidal output current. By Fourier analysis, we know that a non-sinusoidal periodic waveform consists of the fundamental sinusoidal component (whose frequency is same as the frequency of non-sinusoidal periodic waveform) and its harmonics. When this current is passed through the *LC* circuit, which is tuned to the fundamental frequency, it produces the full amplified fundamental signal at the output. Observe that this is a single frequency amplifier depending on the values of L and C. The resonant frequency is given by

$$f = \frac{1}{2\pi\sqrt{LC}} \tag{7.1}$$

The values of L and C are selected such that the resonant frequency is equal to the fundamental frequency.

Class C amplifiers are thus used only in communication circuits, where frequency selection is important. The negative bias is applied through a radio frequency coil to isolate the high-frequency input signal and the dc bias source.These discussions are summarized in Table 7.1.

Table 7.1 Operating cycle of various power-amplifiers

Parameter \ Class	*A*	*B*	*AB*	*C*
Operating cycle	360°	180°	180°	< 180°

7.2 CLASS A POWER AMPLIFIER

Depending on how the load is connected at the amplifier output, we have two types of class A power amplifiers as given below.

1. Series-fed Class A power amplifier.
2. Transformer-coupled Class A power amplifier.

7.2.1 Series-fed Class A Power Amplifier

A fixed-bias series-fed class A power amplifier is shown in Fig. 7.6.

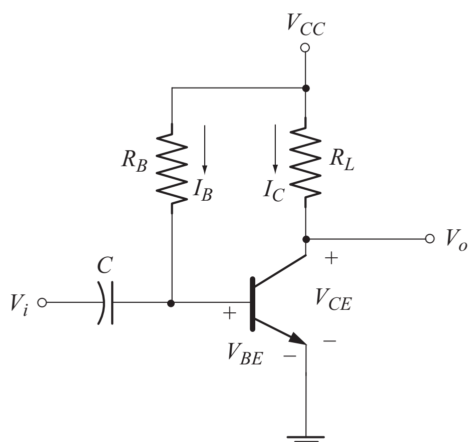


Fig. 7.6 Series-fed class A power amplifier

This circuit is called a series-fed amplifier because the load R_L is connected in series with the collector.

Since R_L is in the collector circuit, it can as well be denoted by R_C.

DC Analysis

From Fig. 7.6, the base current

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} \tag{7.2}$$

and the collector current

$$I_C = \beta I_B \tag{7.3}$$

where β is the dc current gain of the transistor in the CE configuration.

Applying KVL to the collector-emitter circuit, we have

$$V_{CC} = I_C R_L + V_{CE} \tag{7.4}$$

$$\therefore \qquad I_C = \frac{V_{CC} - V_{CE}}{R_L} \tag{7.5}$$

Equation (7.5) can be written as

$$I_C = \left(-\frac{1}{R_L}\right) V_{CE} + \frac{V_{CC}}{R_L} \tag{7.6}$$

where the equation of the dc load is in the slope-intercept form.

Thus, the slope of the dc loadline is $\left(-\frac{1}{R_L}\right)$ and the intercept on the current axis is $\frac{V_{CC}}{R_L}$ as shown in Fig. 7.7.

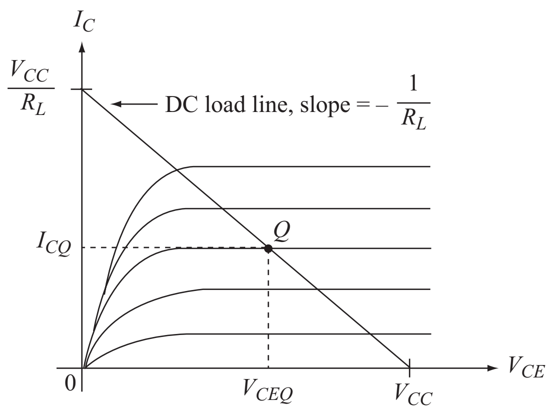


Fig. 7.7 DC load line of class A series-fed power amplifier

AC Analysis

In the circuit of Fig. 7.6, both the dc current and the ac current flows through the same load R_L connected in series with the collector. Hence, the ac load line is the same as the dc load line. The output voltage and current excursions of the series-fed Class A power amplifier is shown in Fig. 7.8.

Observe in Fig. 7.8 that as the amplitude of the input signal increases, the amplitude of the output current as well as the output voltage increases to a maximum extent bounded by

$$0 \text{ and } \frac{V_{CC}}{R_L} \text{ for output current}$$

and

$$0 \text{ and } V_{CC} \text{ for output voltage.}$$

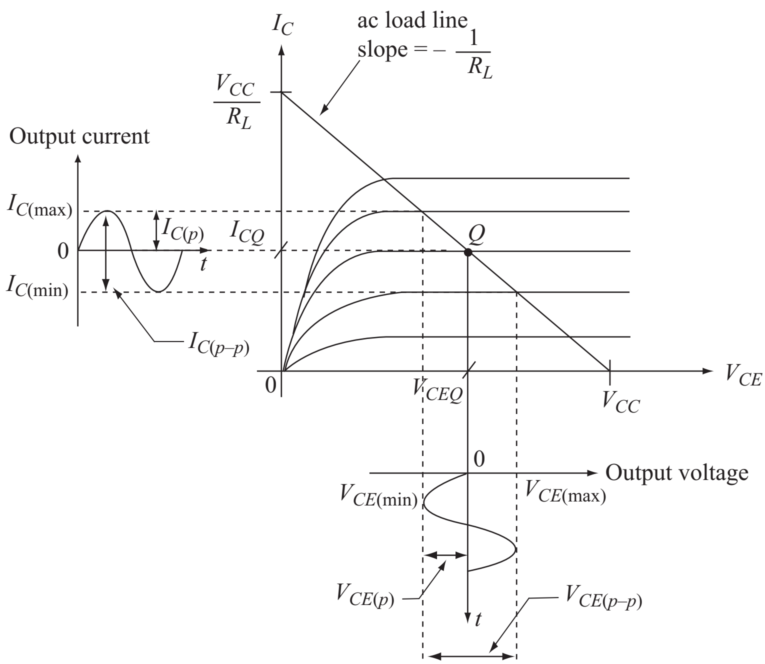


Fig. 7.8 Output voltage and current excursions of the series-fed class A power amplifier

Power Considerations

For the power amplifier, the input power is supplied from the dc source V_{CC}.

∴ The dc power input is given by

$$P_{i(dc)} = V_{CC}\, I_{CQ}$$

The dc power $P_{i(dc)}$, will be continuously drawn from the supply V_{CC} regardless of whether the ac input signal is present or not. The circuit operates with low efficiency due to continuous power dissipation in the collector circuit.

The ac output power delivered to the load can be expressed in the following ways:

Using RMS Values

The ac power delivered to the load is given by

$$P_{o(\text{ac})} = V_{CE(\text{rms})} I_{C(\text{rms})} \tag{7.7}$$

But $$I_{C(\text{rms})} = \frac{V_{CE(\text{rms})}}{R_L} \text{ or } V_{CE(\text{rms})} = I_{C\,(\text{rms})} R_L$$

Using these relations in Equation (7.7), the ac output power can be expressed as

$$P_{o(\text{ac})} = \frac{V^2_{CE(\text{rms})}}{R_L} \tag{7.8}$$

or $$P_{o(\text{ac})} = I^2_{C(\text{rms})} R_L \tag{7.9}$$

Using Peak Values

We know that, $$V_{CE(\text{rms})} = \frac{V_{CE(p)}}{\sqrt{2}} \text{ and } I_{C\,(\text{rms})} = \frac{I_{C(p)}}{\sqrt{2}}$$

where $V_{CE(p)}$ = peak load voltage
and $I_{C(p)}$ = peak load current

Using these relations in Equation (7.7) we have

$$P_{o(\text{ac})} = \frac{V_{CE(p)}}{\sqrt{2}} \cdot \frac{I_{C(p)}}{\sqrt{2}}$$

$$= \frac{V_{CE(p)} I_{C(p)}}{2} \tag{7.10}$$

But $$V_{CE\,(p)} = I_{C\,(p)} R_L \Rightarrow I_{C\,(p)} = \frac{V_{CE(p)}}{R_L}$$

Using these relations in Equation (7.10) we have

$$P_{o(\text{ac})} = \frac{V^2_{CE(p)}}{2R_L} \tag{7.11}$$

or $$P_{o(\text{ac})} = \frac{I^2_{C(p)}}{2} R_L \tag{7.12}$$

Using Peak-to-peak Values

The peak-to-peak load voltage is

$$V_{CE(p-p)} = 2\,V_{CE(p)} \Rightarrow V_{CE(p)} = \frac{V_{CE(p-p)}}{2}$$

and the peak-to-peak load current is

$$I_{C(p-p)} = 2\, I_{C(p)} \quad \Rightarrow \quad I_{C(p)} = \frac{I_{C(p-p)}}{2}$$

Using these relations in Equation (7.10) we have

$$P_{o(\text{ac})} = \frac{V_{CE(p-p)}\, I_{C(p-p)}}{8} \tag{7.13}$$

Equation (7.11) can be rewritten as

$$P_{o(\text{ac})} = \frac{V^2_{CE(p-p)}}{8\, R_L} \tag{7.14}$$

and from Equation (7.12) we have

$$P_{o(\text{ac})} = \frac{I^2_{C(p-p)}}{8}\, R_L \tag{7.15}$$

Using Maximum and Minimum Values

From Fig. 7.8

$$V_{CE(p-p)} = V_{CE(\max)} - V_{CE(\min)} \tag{7.16}$$

$$I_{C(p-p)} = I_{C(\max)} - I_{C(\min)} \tag{7.17}$$

Using these relations in Equation (7.13) we have

$$P_{o(\text{ac})} = \frac{[V_{CE(\max)} - V_{CE(\min)}][I_{C(\max)} - I_{C(\min)}]}{8} \tag{7.18}$$

Maximum ac Output Power

From Equation (7.18)

$P_{o(\text{ac})}$ is maximum, when

$$\left.\begin{aligned} V_{CE(\min)} &= 0 \qquad & I_{C(\min)} &= 0 \\ V_{CE(\max)} &= V_{CC} \qquad & I_{C(\max)} &= \frac{V_{CC}}{R_L} \end{aligned}\right\} \tag{7.19}$$

Substituting in Equation (7.18)

$$P_{o(\text{ac})\max} = \frac{V_{CC}}{8} \cdot \frac{V_{CC}}{R_L}$$

or

$$P_{o(\text{ac})\max} = \frac{V^2{}_{CC}}{8R_L} \tag{7.20}$$

This equation can be used to select the value of V_{CC} for a desired power output and a given load.

Note that in the analysis, we have assumed that the transistor is linear and thus both the input and output signals are purely sinusoidal.

The merits of the amplifier are that it is simple and no coupling element is required. The serious **drawbacks** however are

- There is an impedance mismatch between the power amplifier and the load.
- It has very poor conversion efficiency (see section 7.5.1)
- Most often, the load for power amplifier is the loud speaker which is basically a magnetic circuit. In the series-fed amplifier, both ac and dc current would flow through this magnetic load resulting in core saturation.

Example 7.1

A class A series-fed power amplifier is required to deliver a maximum power of 20 W to a load of 4 Ω. Calculate the required supply voltage.

Solution

Given,

$$P_{o(\text{ac})\max} = 20\text{ W} \quad R_L = 4\ \Omega$$

From Equation (7.20)

$$P_{o(\text{ac})\max} = \frac{V^2{}_{CC}}{8R_L}$$

$$\therefore \qquad V_{CC} = \sqrt{8P_{o(\text{ac})\max}\,R_L}$$

$$= \sqrt{8\times 20\times 4} = \sqrt{640} = 25.29\text{ V}$$

$$V_{CC} = 25.29\text{ V}$$

7.2.2 Transformer-Coupled Class A Power Amplifier

This is also referred to often as the transformer-coupled audio power amplifier. We saw that one of the serious drawbacks of the series-fed power amplifier was impedance mismatch.

Impedance Matching using Transformer

Consider the output equivalent circuit of the series fed class A power amplifier shown in Fig. 7.9.

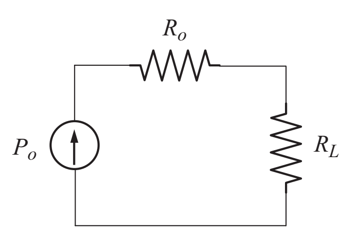


Fig. 7.9 Output equivalent circuit of series-fed class A power amplifier

P_o represents the ac power developed at the output of power amplifier and R_o is its output resistance.

For transistor amplifier, $R_o \approx \dfrac{1}{h_{oe}}$ which is of the order of kilo ohms. But the typical load for an audio power amplifier is loud speaker whose impedance is of the order of 5 to 50 Ω.

We know from the maximum power transfer theorem that maximum power is transferred to the load when $R_L = R_o$. This is not the case in audio amplifiers, resulting in a reduced power being delivered to the load because of impedance mismatch. For maximum power transfer, the value of R_L needs to be boosted to the value of R_o. This is called impedance transformation which can be done using a transformer. Impedance transformation can be explained with Fig. 7.10 .

In Fig. 7.10,

V_1 = primary voltage
V_2 = secondary voltage
I_1 = primary current
I_2 = secondary current

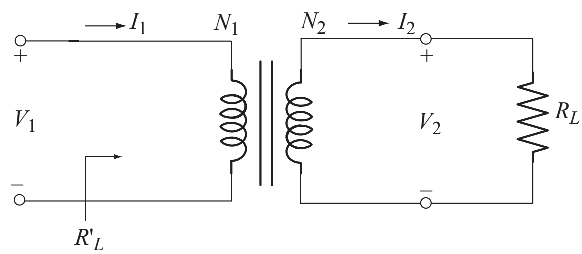


Fig. 7.10 Impedance transformation using transformer

Load on the secondary,

$$R_L = \frac{V_2}{I_2} \tag{7.21}$$

$$R_L' = \frac{V_1}{I_1} \tag{7.22}$$

where R_L' is the load reflected at the primary.

For the transformer

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} \tag{7.23}$$

and

$$\frac{I_2}{I_1} = \frac{N_1}{N_2} \tag{7.24}$$

Multiplying Equations (7.23) and (7.24)

$$\frac{V_1 I_2}{V_2 I_1} = \frac{N_1^2}{N_2^2} \tag{7.25}$$

$$\frac{V_1/I_1}{V_2/I_2} = \left(\frac{N_1}{N_2}\right)^2 \tag{7.26}$$

Substituting Equations (7.21) and (7.22) in Equation (7.26), we have

$$\frac{R_L'}{R_L} = \left(\frac{N_1}{N_2}\right)^2$$

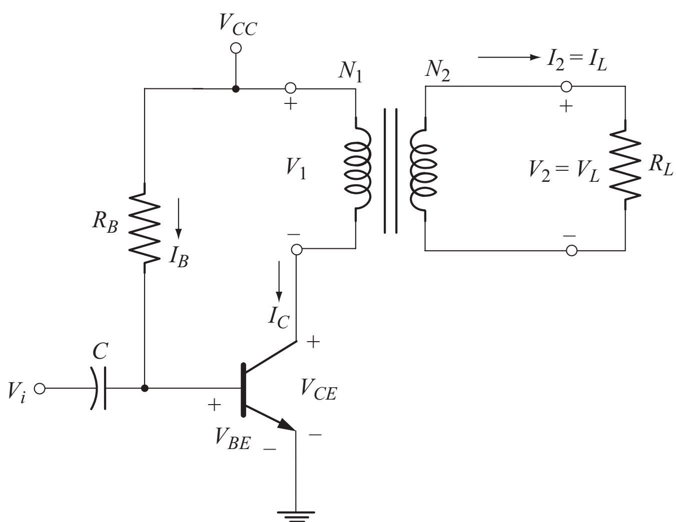
or

$$R_L' = R_L\left(\frac{N_1}{N_2}\right)^2 \tag{7.27}$$

$R_L' > R_L$ can be achieved by choosing $N_1 > N_2$. i.e., we have to use a step down transformer of appropriate turns ratio.

Analysis of Amplifier Stage

A transformer-coupled class A power amplifier is shown in Fig. 7.11(a).



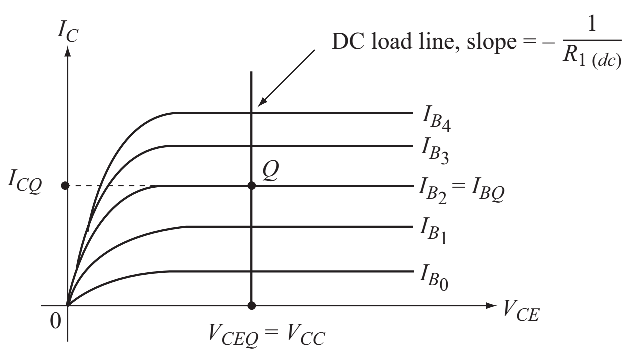


Fig. 7.11 Transformer-coupled power amplifier
(a) Circuit (b) DC load line

DC Analysis

Applying KVL to the collector- emitter circuit of Fig. 7.11 (a), we have,

$$V_{CC} = I_C R_{1\,(dc)} + V_{CE} \tag{7.28}$$

where $R_{1\,(dc)}$ is the dc resistance of the primary winding of the transformer.

Equation (7.28) can be re-written as

$$I_C R_{1\,(dc)} = V_{CC} - V_{CE} \tag{7.29}$$

$$\therefore \qquad I_C = \left(-\frac{1}{R_{1\,(dc)}}\right) V_{CE} + \frac{V_{CC}}{R_{1\,(dc)}} \tag{7.30}$$

which is the equation of the dc load line in the slope-intercept form, with slope $= -\dfrac{1}{R_{1\,(dc)}}$ and current-axis intercept $\dfrac{V_{CC}}{R_{1\,(dc)}}$. Since, the dc resistance of the transformer will be very low of the order of a few ohms, the slope is close to ∞ and the dc load line will be almost vertical as shown in Fig. 7.11(b).

Observe in Equation (7.28) that $I_C R_{1\,(dc)}$ will be close to zero and therefore $V_{CE} = V_{CC}$.

The intersection of the dc load line and the base current set by the biasing circuit defines the operating point as illustrated in Fig. 7.11(b) for $I_{B_2} = I_{BQ}$.

AC Analysis

The ac load line has a slope of $-\dfrac{1}{R'_L}$, passing through the Q-point as shown in Fig. 7.12.

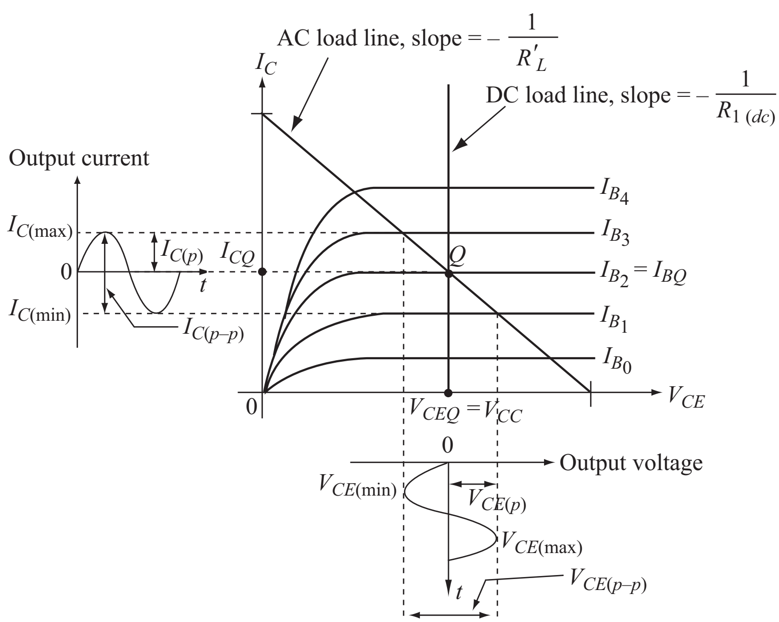


Fig. 7.12 The ac load line with output voltage and current variations in a transformer-coupled class A power amplifier

From Fig. 7.12, it is clear that the maximum output voltage at the primary will exceed V_{CC}. Thus, it is necessary to ensure that the possible voltage swings do not exceed the transistor voltage ratings.

The dc power drawn by the circuit is given by

$$P_{i(dc)} = V_{CC} I_{CQ} \tag{7.31}$$

Let us assume that the efficiency of the transformer is 100 percent so that the ac power developed in the collector circuit or the primary of the transformer equals the ac power delivered to the load on the secondary side.

From Equation (7.10), the ac power developed across the primary is given by

$$P_{o(ac)} = \frac{V_{CE(p)} I_{C(p)}}{2} \tag{7.32}$$

From Fig. 7.12,

$$V_{CE(max)} = V_{CC} + V_{CE(p)} \tag{7.33}$$

and $$V_{CE(min)} = V_{CC} - V_{CE(p)} \tag{7.34}$$

Adding Equations (7.33) and (7.34) we get

$$V_{CE(max)} + V_{CE(min)} = 2\,V_{CC}$$

or $$V_{CC} = \frac{V_{CE(max)} + V_{CE(min)}}{2} \tag{7.35}$$

Subtracting Equations (7.33) and (7.34), we get

$$V_{CE(max)} - V_{CE(min)} = 2\,V_{CE(p)} = V_{CE(p\text{-}p)}$$

or $$V_{CE(p)} = \frac{V_{CE(max)} - V_{CE(min)}}{2} = \frac{V_{CE(p-p)}}{2} \tag{7.36}$$

similarly $$I_{C(p)} = \frac{I_{C(max)} - I_{C(min)}}{2} = \frac{I_{C(p-p)}}{2} \tag{7.37}$$

Using these relations in Equation (7.32) we get

$$P_{o(ac)} = \frac{[V_{CE(max)} - V_{CE(min)}][I_{C(max)} - I_{C(min)}]}{8} \tag{7.38}$$

or $$P_{o(ac)} = \frac{V_{CE(p-p)} I_{C(p-p)}}{8} \tag{7.39}$$

But $$I_{C(p\text{-}p)} = \frac{V_{CE(p-p)}}{R'_L} \Rightarrow V_{CE(p\text{-}p)} = I_{C(p\text{-}p)} R'_L$$

Using these relations in Equation (7.39) we have

$$P_{o(ac)} = \frac{V^2_{CE(p-p)}}{8R'_L} = \frac{V^2_{CE(p)}}{2R'_L} \tag{7.40}$$

$$\text{or} \qquad P_{o(\text{ac})} = \frac{I^2_{C(p-p)}}{8} R'_L = \frac{I^2_{C(p)}}{2} R'_L \tag{7.41}$$

AC Power Delivered to the Load

The ac power delivered to the load on the secondary side is

$$P_L = V_{L(\text{rms})} I_{L(\text{rms})} \tag{7.42}$$

$$\text{Using} \qquad I_{L(\text{rms})} = \frac{V_{L(\text{rms})}}{R_L}$$

$$P_L = \frac{V^2_{L(\text{rms})}}{R_L} \tag{7.43}$$

$$\text{or} \qquad P_L = I^2_{L\,(\text{rms})} R_L \tag{7.44}$$

The load current $I_{L(\text{rms})}$ can be obtained from

$$\frac{I_{L(\text{rms})}}{I_{C(\text{rms})}} = \frac{N_1}{N_2} \qquad [\because I_{L(\text{rms})} = I_{2(\text{rms})}]$$

$$\text{or} \qquad I_{L(\text{rms})} = \frac{N_1}{N_2} I_{C\,(\text{rms})} \tag{7.45}$$

Since we have assumed ideal transformer for which the efficiency is 100 percent,

$$P_L = P_{o(\text{ac})} \tag{7.46}$$

If the transformer is not ideal, P_L and $P_{o(\text{ac})}$ are related by

$$P_L = \eta_{\text{TFR}} P_{o(\text{ac})} \tag{7.47}$$

where η_{TFR} is the efficiency of the transformer.

Maximum ac Output Power

From Equation (7.38), we find that the ac output power $P_{o(\text{ac})}$ is maximum when $V_{CE(\min)} = 0$ and $I_{C(\min)} = 0$

$$\text{Thus,} \qquad P_{o(\text{ac})\max} = \frac{V_{CE(\max)} I_{C(\max)}}{8}$$

$$\text{Using} \qquad I_{C\,(\max)} = \frac{V_{CE(\max)}}{R'_L}, \quad \text{we have}$$

$$P_{o(\text{ac})\max} = \frac{V^2_{CE(\max)}}{8R'_L} \tag{7.48}$$

From Equation (7.35),

$$\text{with} \qquad V_{CE\,(\min)} = 0, \quad \text{we get}$$

$$V_{CE(\max)} = 2\,V_{CC}$$

Using this relation in Equation (7.48) we have

$$P_{o(\text{ac})\max} = \frac{V_{CC}^2}{2R'_L} \tag{7.49}$$

Example 7.2

A class A transformer coupled audio power amplifier is requird to deliver a maximum of 1 W into a loud speaker of 10 Ω resistance. If the output resistance of the amplifier is 1000 Ω, calculate

(a) Turns ration of the transformer required
(b) Power supply voltage
Assume an ideal transformer (100 % efficiency)

Solution

Given

$$P_{o(\text{ac})\max} = 1\text{ W} \quad R_L = 10\ \Omega \quad R_o = 1000\ \Omega$$

(a) From Equation (7.27)

$$R'_L = R_L\left(\frac{N_1}{N_2}\right)^2$$

$$\therefore \quad \left(\frac{N_1}{N_2}\right)^2 = \frac{R'_L}{R_L}$$

For maximum power transfer,

$$R'_L = R_o$$

$$\therefore \quad \frac{N_1}{N_2} = \sqrt{\frac{R_o}{R_L}} = \sqrt{\frac{1000}{10}} = 10$$

or $\quad N_1 : N_2$ is 10:1

A 10:1 step down transformer is to be used.

(b) From Equation (7.49)

$$P_{o\,(\text{ac})\max} = \frac{V_{CC}^2}{2R'_L} = \frac{V_{CC}^2}{2R_o}$$

$$\therefore \quad V_{CC} = \sqrt{2R_o\,P_{o(\text{ac})\max}}$$

$$= \sqrt{2 \times 1000 \times 1}$$

$$V_{CC} = 44.7\text{ V}$$

Let $\quad V_{CC} = 45$ V

Example 7.3

Repeat Example 7.2, if the transformer efficiency is 75%.

Solution

Given,

$$P_{o(\text{ac})\max} = 1\text{W} \quad R_L = 10\ \Omega \quad R_o = 1000\ \Omega \quad \text{Transformer efficiency} = 75\ \%$$

(a) Transformer turns ratio is 10 as before.

(b) $P_{o(\text{ac})\max}$ represents the maximum ac power developed in the primary of the transformer.

It is required to deliver 1 W of power in the load connected to the secondary.

$$\text{Therefore, } P_{o(\text{ac})\max} = \frac{\text{Power delivered to } R_L}{\text{Transformer efficiency}} = \frac{1\text{W}}{0.75} = \frac{4}{3}\ \text{W}$$

$$\therefore \qquad V_{CC} = \sqrt{2R_o P_{o(\text{ac})\max}} = \sqrt{2\times 1000\times\frac{4}{3}} = 51.64\ \text{V}$$

$$\text{Let} \qquad V_{CC} = 52\ \text{V}.$$

Observe that a higher supply voltage is required if the transformer is not ideal.

Example 7.4

A load of $10\ \Omega$ is connected between the secondary terminals of a 20:1 transformer. Calculate the effective resistance seen looking into the primary terminals.

Solution

$$R_L' = \left(\frac{N_1}{N_2}\right)^2 R_L$$

$$= (20)^2\ (10\ \Omega)$$

$$= 4\ \text{k}\Omega$$

Example 7.5

Find the turns ratio of the transformer required to match a $6\ \Omega$ speaker load so that the effective load resistance seen at the primary is $8\ \text{k}\Omega$?

Solution

$$R_L' = \left(\frac{N_1}{N_2}\right)^2 R_L$$

$$\frac{N_1}{N_2} = \sqrt{\frac{R_L'}{R_L}} = \sqrt{\frac{8\text{k}\Omega}{6\Omega}} = 36.51$$

7.3 HARMONIC DISTORTION OR NON LINEAR DISTORTION

So far we have assumed that the transistor is linear. However, the dynamic transfer characteristic is non linear as shown in Fig (a). The non-linearity arises because the static output characteristics are not equidistant straight lines for constant increments in the input excitation. If the dynamic transfer curve is nonlinear over the signal excursion range, the output will not be sinusoidal when the input is sinusoidal. This type of distortion is called non-linear distortion or harmonic distortion.

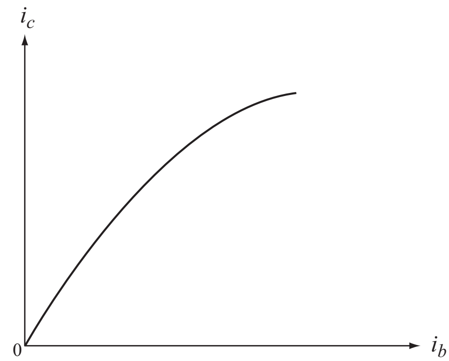


Fig. (a) Dynamic transfer characteristics of a transistor

In the presence of non linearity, the input-output relation can be expressed as

$$i_c = G_1 i_b + G_2 i_b^2 + G_3 i_b^3 + \cdots \tag{A}$$

where the first term represents the linear term followed by the non-linear terms. For instance if the input is sinusoidal of large amplitude, say

$$i_b = I_{bm} \sin \omega t \tag{B}$$

Then Equation (A) becomes

$$i_c = G_1 I_{bm} \sin\omega t + G_2 I_{bm}^2 \sin^2 \omega t + G_3 I_{bm}^3 \sin^3 \omega t + \cdots$$

which can be expressed in the general form as

$$i_c = B_0 + B_1 \sin\omega t + B_2 \sin 2\omega t + B_3 \sin 3\omega t + \cdots \tag{C}$$

In Equation (C)

- B_0 is the dc term
- $B_1 \sin \omega t$ is the fundamental component whose frequency is same as the input frequency (i.e ω)
- $B_2 \sin 2\omega t$ is the second harmonic component whose frequency is twice the input frequency (i.e. 2ω)
- $B_3 \sin 3\omega t$ is the third harmonic component whose frequency is thrice the input frequency (i.e. 3ω) and so on.

ω is called the fundamental frequency, 2 ω the second harmonic frequency, 3 ω the third harmonic frequency and so on.

Since the non-linearity of the dynamic transfer characteristics of the transistor results in the generation of harmonics of the input frequency, the distortion is termed non-linear distortion or harmonic distortion.

The Fourier analysis of the output signal reveals that as the order of the harmonic increases, its amplitude decreases

i.e., $$|B_1| > |B_2| > |B_3| > \cdots \tag{D}$$

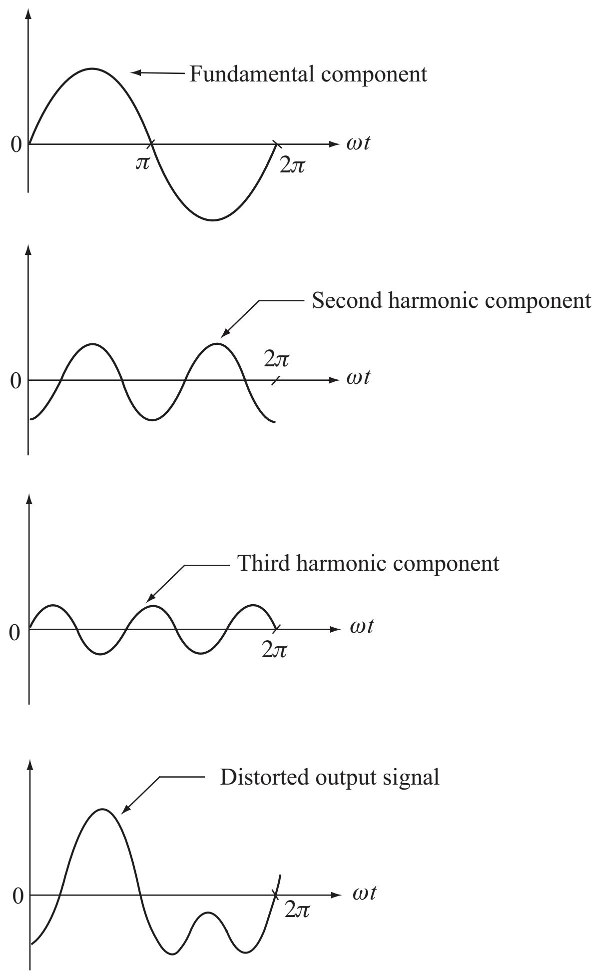


Fig. (b) *Graphical representation of harmonic components and the distorted output signal*

From the above equation we find that the second harmonic component has the largest amplitude of all other harmonic components and it is the principal source of harmonic distortion.

Fig. (b) shows the graphical description of Harmonic components of a distorted signal.

7.3.1 Second Harmonic Distortion

In the previous section we found that the second harmonic component has the largest amplitude of all other harmonic components which is the principal source of harmonic distortion. Now Let us proceed to estimate the magnitude of second harmonic distortion by considering the second order non linearity.

The relationship between the collector current and base current is given by

$$i_c = G_1\, i_b + G_2\, i_b^2 \tag{7.50}$$

where, i_c and i_b represent the instantaneous collector and base current respectively and G_1 and G_2 are constants.

Let the input base current be of the form

$$i_b = I_{bm} \cos \omega t \tag{7.51}$$

i_b is represented by $I_{bm} \cos \omega t$ for sake of computational simplicity; it could as well be $I_{bm} \sin \omega t$ where, I_{bm} is peak value of base current. Substituting Equation (7.51) in Equation (7.50)

$$i_c = G_1\,(I_{bm} \cos \omega t) + G_2\,(I_{bm} \cos \omega t)^2 \tag{7.52}$$

But $$\cos^2 \omega t = \frac{1 + \cos 2\,\omega t}{2} \tag{7.53}$$

$$\therefore \quad i_c = G_1\,(I_{bm} \cos \omega t) + G_2\, I_{bm}^2 \frac{(1 + \cos 2\,\omega t)}{2}$$

$$= G_1\,(I_{bm} \cos \omega t) + \frac{G_2 I_{bm}^2}{2} + \frac{G_2 I_{bm}^2}{2} \cos 2\,\omega t$$

$$i_c = \frac{G_2 I_{bm}^2}{2} + G_1\, I_{bm} \cos \omega t + \frac{G_2 I_{bm}^2}{2} \cos 2\,\omega t \tag{7.54}$$

Let $$B_0 = \frac{G_2 I_{bm}^2}{2} = B_2$$

and $$B_1 = G_1\, I_{bm}$$

where B_0 represents the dc term while B_1 represents the peak value of the fundamental output current and B_2 the peak value of the second harmonic component of the output current.

Now Equation (7.54) becomes

$$i_c = B_0 + B_1 \cos \omega t + B_2 \cos 2\,\omega t \tag{7.55}$$

From Equation (7.55) observe that under second order non-linearity, we get

- B_0, a dc term indicating that a part of the input signal is being rectified.
- $B_1 \cos \omega t$, an ac term which represents the amplified input signal and
- $B_2 \cos 2\,\omega t$, another ac term which represents the second harmonic component.

Thus, because of the dc term and the second harmonic component, the output signal will not be an exact replica of the input signal.

Now, the total instantaneous collector current

$$i_C = \text{Quiescent collector current} + \text{instantaneous collector current}$$

$$i_C = I_{CQ} + i_c \tag{7.56}$$

Substituting for i_c from Equation (7.55), we have

$$i_C = I_{CQ} + B_0 + B_1 \cos \omega t + B_2 \cos 2\,\omega t \tag{7.57}$$

We can estimate the amplitudes of B_0, B_1 and B_2 from the dynamic transfer characteristics shown in Fig. 7.13.

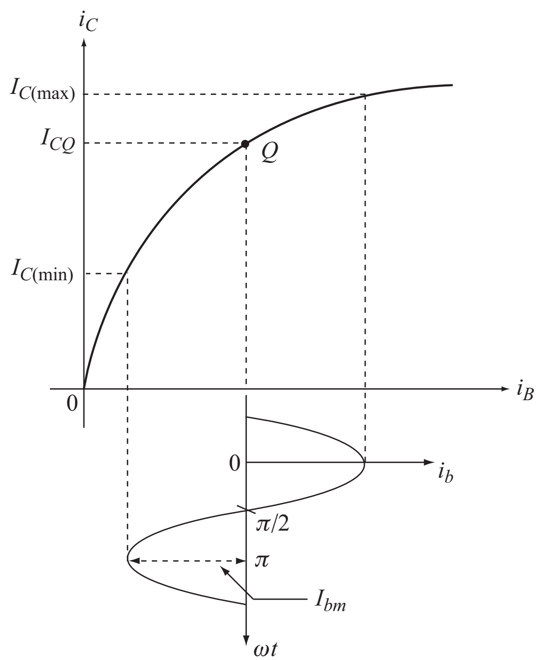


Fig. 7.13 Computation of harmonic components

There are 3 unknowns in Equation (7.57), i.e., B_0, B_1 and B_2 and hence we consider three points, i.e, $I_{C(\max)}$, I_{CQ} and $I_{C(\min)}$ on the dynamic transfer characteristics. This procedure is known as the **3-point method** of calculating harmonic distortion.

From Fig. 7.13, we have,

$$\left.\begin{aligned} &\text{at } \omega t = 0, && i_C = I_{C(\max)} \\ \text{and} \quad &\text{at } \omega t = \frac{\pi}{2}, && i_C = I_{CQ} \\ &\text{at } \omega t = \pi, && i_C = I_{C(\min)} \end{aligned}\right\} \tag{7.58}$$

Applying this condition to Equation (7.57), we get

$$I_{C(\max)} = I_{CQ} + B_0 + B_1 + B_2 \tag{7.59}$$

$$I_{CQ} = I_{CQ} + B_0 - B_2$$

$$\Rightarrow \quad B_0 = B_2 \tag{7.60}$$

as expected, and

$$I_{C(\min)} = I_{CQ} + B_0 - B_1 + B_2 \tag{7.61}$$

Applying Equation (7.60) to Equations (7.59) and (7.61),

$$I_{C(\max)} = I_{CQ} + 2\,B_0 + B_1 \tag{7.62}$$

$$\text{and} \quad I_{C(\min)} = I_{CQ} + 2\,B_0 - B_1 \tag{7.63}$$

Subtracting Equation (7.63) from Equation (7.62)

$$I_{C(\max)} - I_{C(\min)} = 2\,B_1$$

$$\text{or} \quad B_1 = \frac{I_{C(\max)} - I_{C(\min)}}{2} \tag{7.64}$$

Adding equations (7.62) and (7.63)

$$I_{C(\max)} + I_{C(\min)} = 2\,I_{CQ} + 4\,B_0$$

$$\therefore \quad B_0 = B_2 = \frac{I_{C(\max)} + I_{C(\min)} - 2\,I_{CQ}}{4} \tag{7.65}$$

% second harmonic distortion is given by

$$D_2 = \frac{\text{Magnitude of second harmonic component}}{\text{Magnitude of fundamental}} \times 100\%$$

$$D_2 = \frac{|B_2|}{|B_1|} \times 100\% \tag{7.66}$$

Substituting for B_1 and B_2 we get

$$D_2 = \left| \frac{\frac{1}{2}[I_{C(\max)} + I_{C(\min)}] - I_{CQ}}{I_{C\max} - I_{C\min}} \right| \times 100\% \tag{7.67}$$

In a similar manner, the second harmonic distortion can expressed interms of collector-emitter voltages as

$$D_2 = \left| \frac{\frac{1}{2}[V_{CE(\max)} + V_{CE\,(\min)}] - V_{CEQ}}{V_{CE(\max)} - V_{CE(\min)}} \right| \times 100\% \tag{7.68}$$

Example 7.6

The following readings are available for a power amplifier. Calculate the second harmonic distortion in each case.

(a) $V_{CEQ} = 10$ V $\quad V_{CE(\max)} = 18$ V $\quad V_{CE(\min)} = 1$ V

(b) $V_{CEQ} = 10$ V $\quad V_{CE(\max)} = 19$ V $\quad V_{CE(\min)} = 1$ V

Solution

(a)

$$D_2 = \left| \frac{\frac{1}{2}[V_{CE(\max)} + V_{CE(\min)}] - V_{CEQ}}{V_{CE(\max)} - V_{CE(\min)}} \right| \times 100\%$$

$$= \left| \frac{\frac{1}{2}[18\text{V} + 1\text{V}] - 10\text{V}}{18\text{V} - 1\text{V}} \right| \times 100\% = 2.94\%$$

(b)

$$D_2 = \left| \frac{\frac{1}{2}[19\text{V} + 1\text{V}] - 10\text{V}}{19\text{V} - 1\text{V}} \right| \times 100\% = 0\%$$

Example 7.7

A transistor supplies 0.85 W to a 4 kΩ load. The zero signal DC collector current is 31 mA and the dc collector current with signal is 34 mA. Determine the second harmonic distortion.

Solution

Given,

$$P_{o(ac)} = 0.85 \text{ W} \qquad R_L = 4 \text{ k}\Omega$$

Under zero signal, the collector current is the quiscent collector current.

$$\therefore \quad I_{CQ} = 31 \text{ mA}$$

When the signal is present, the dc collector current is $I_{CQ} + B_0$

$$\therefore \quad I_{CQ} + B_0 = 34 \text{ mA}$$

$$\therefore \quad B_0 = 34 \text{ mA} - 31 \text{ mA} = 3 \text{ mA}$$

$$\therefore \quad B_0 = B_2 = 3 \text{ mA}$$

The given $P_{o(ac)}$ is the power delivered by the fundamental component.

$$P_{o(ac)} = \left(\frac{I_{L(p)}}{\sqrt{2}} \right)^2 \times R_L$$

Since B_1 is the peak value of the fundamental, $I_{L(p)} = B_1$

$$\therefore \quad P_{o(\text{ac})} = \left(\frac{B_1}{\sqrt{2}}\right)^2 \times R_L$$

$$\frac{B_1^2}{2} = \frac{P_{o(\text{ac})}}{R_L}$$

$$B_1 = \sqrt{\frac{2\,P_{o(\text{ac})}}{R_L}} = \sqrt{\frac{2 \times 0.8\text{ W}}{4\text{k}\Omega}} = 20\text{ mA}$$

$$\therefore \quad \%\,D_2 = \frac{|B_2|}{|B_1|} \times 100\%$$

$$= \frac{3\text{ mA}}{20\text{ mA}} \times 100\% = 15\%$$

Example 7.8

Non-linear distortion results in the generation of frequencies in the output that are not present in the input. If the dynamic curve can be represented by the equation $i_c = G_1\, i_b + G_2\, i_b^2$ and if the input signal is given by $i_b = (I_1 \cos\omega_1 t + I_2 \cos\omega_2 t)$, show that the output will contain a DC term and sinusoidal terms of frequencies $\omega_1, \omega_2, 2\,\omega_1, 2\,\omega_2, (\omega_1 + \omega_2)$ and $(\omega_1 - \omega_2)$.

Solution

Given

$$i_c = G_1\, i_b + G_2\, i_b^2 \tag{A}$$

and

$$i_b = I_1 \cos\omega_1 t + I_2 \cos\omega_2 t \tag{B}$$

Substituting Equation (B) into (A)

$$i_c = G_1\,(I_1 \cos\omega_1 t + I_2 \cos\omega_2 t) + G_2\,(I_1 \cos\omega_1 t + I_2 \cos\omega_2 t)^2$$

$$i_c = G_1\, I_1 \cos\omega_1 t + G_1\, I_2 \cos\omega_2 t + G_2\,(I_1^2 \cos^2\omega_1 t + I_2^2 \cos^2\omega_2 t + 2\, I_1\, I_2 \cos\omega_1 t \cos\omega_2 t) \tag{C}$$

$$\cos^2\omega_1 t = \frac{1 + \cos 2\,\omega_1 t}{2}$$

$$\cos^2\omega_2 t = \frac{1 + \cos 2\,\omega_2 t}{2}$$

$$\cos\omega_1 t\;\cos\omega_2 t = \frac{\cos(\omega_1 + \omega_2)t + \cos(\omega_1 - \omega_2)t}{2}$$

Using these relations in Equation (C)

$$i_c = G_1\, I_1 \cos\omega_1 t + G_1\, I_2 \cos\omega_2 t + G_2\, I_1^2\, \frac{(1 + \cos 2\omega_1 t)}{2} + G_2\, I_2^2\, \frac{(1 + \cos 2\omega_2 t)}{2} +$$

$$2\, G_2\, I_1\, I_2 \left(\frac{\cos(\omega_1 + \omega_2)t + \cos(\omega_1 - \omega_2)t}{2}\right)$$

$$= G_1 I_1 \cos \omega_1 t + G_1 I_2 \cos \omega_2 t + \frac{G_2 I_1^2}{2} + \frac{G_2 I_1^2}{2} \cos 2\omega_1 t + \frac{G_2 I_2^2}{2}$$

$$+ \frac{G_2 I_2^2}{2} \cos 2\omega_2 t + G_2 I_1 I_2 \cos(\omega_1 + \omega_2)\, t + G_2 I_1 I_2 \cos(\omega_1 - \omega_2)\, t$$

$$= \frac{G_2}{2} (I_1^2 + I_2^2) + G_1 I_1 \cos \omega_1 t + G_1 I_2 \cos \omega_2 t + \frac{G_2 I_1^2}{2} \cos 2\omega_1 t$$

$$+ \frac{G_2 I_2^2}{2} \cos 2\omega_2 t + G_2 I_1 I_2 \cos(\omega_1 + \omega_2)\, t + G_2 I_1 I_2 \cos(\omega_1 - \omega_2)\, t \qquad \text{(D)}$$

Observe that Equation (D) has a dc component besides the cosine terms with frequencies ω_1, ω_2, $2\omega_1$, $2\omega_2$, $(\omega_1 + \omega_2)$ and $(\omega_1 - \omega_2)$

7.4 HIGHER-ORDER HARMONIC DISTORTION

In section 7.3.1, we have considered only second-order non-linearity. In a power amplifier since the magnitudes of voltages and currents will be large, it is necessary to consider higher-order non-linearities also.

We can express the output current by a power series of the form

$$i_c = G_1 i_b + G_2\, i_b^2 + G_3 i_b^3 + G_4 i_b^4 + \cdots + \cdots \qquad (7.69)$$

$$i_b = I_{bm} \cos \omega t$$

Now Equation (7.69) can be written as

$$I_c = G_1(I_{bm} \cos \omega t) + G_2(I_{bm} \cos \omega t)^2 + \mathrm{G}_3\, (I_{bm} \cos \omega t)^3 + \mathrm{G}_4\, (I_{bm} \cos \omega t)^4 + \cdots + \cdots \qquad (7.70)$$

Using proper trigonometric identities, Equation (7.70) can be compactly represented as

$$i_c = B_0 + B_1 \cos \omega t + B_2 \cos 2\, \omega t + B_3 \cos 3\, \omega t + B_4 \cos 4\, \omega t + \cdots \qquad (7.71)$$

Observe the presence of higher-order harmonic terms corresponding to 2ω, 3ω, 4ω etc., in Equation (7.71).

The total instantaneous collector current

$$i_C = I_{CQ} + i_c$$

$$\therefore \qquad i_C = I_{CQ} + B_0 + B_1 \cos \omega t + B_2 \cos 2\, \omega t + B_3 \cos 3\, \omega t + B_4 \cos 4\, \omega t \qquad (7.72)$$

Truncating after the 4th harmonic term.

There are 5 unknowns B_0 to B_4 which can be computed graphically using the ***5-point procedure***.

Calculation of Total Harmonic Distortion

% second harmonic distortion,

$$D_2 = \frac{|B_2|}{|B_1|} \times 100\% \qquad (7.73)$$

% third harmonic distortion

$$D_3 = \frac{|B_3|}{|B_1|} \times 100\% \tag{7.74}$$

and % fourth harmonic distortion is

$$D_4 = \frac{|B_4|}{|B_1|} \times 100\% \tag{7.75}$$

Total harmonic distortion (THD) or distortion factor is

$$\text{THD} = D = \sqrt{D_2^2 + D_3^2 + D_4^2} \tag{7.76}$$

In general

$$D = \sqrt{D_2^2 + D_3^2 + D_4^2 + D_5^2 + \cdots} \tag{7.77}$$

when all the harmonic components are considered.

Now, let us calculate the total power output.

$$P = P_1 + P_2 + P_3 + P_4 + \cdots \tag{7.78}$$

where, P_1 is the power delivered to the load by the fundamental component and
P_i is the power delivered to the load by the i^{th} harmonic component (for $i = 2, 3, \ldots$)
P_1 = (Fundamental rms current)^2 × load resistance

$$\therefore \qquad P_1 = \left(\frac{B_1}{\sqrt{2}}\right)^2 R_L$$

$$P_1 = \left(\frac{B_1^2}{2}\right) R_L \tag{7.79}$$

Similarly for $i = 2, 3, 4 \ldots$

$$P_i = \left(\frac{B_i^2}{2}\right) R_L \tag{7.80}$$

Substituting Equations (7.79) and (7.80) in Equation (7.78), the total power output is given by

$$P = \frac{B_1^2 R_L}{2} + \frac{B_2^2 R_L}{2} + \frac{B_3^2 R_L}{2} + \cdots + \cdots$$

$$P = \frac{B_1^2 R_L}{2} \left[1 + \left(\frac{B_2}{B_1}\right)^2 + \left(\frac{B_3}{B_1}\right)^2 + \cdots\right] \tag{7.81}$$

$$P = P_1 (1 + D_2^2 + D_3^2 + \cdots) \tag{7.82}$$

From Equation (7.77)

$$D^2 = D_2^2 + D_3^2 + D_4^2 + \cdots$$

$\therefore$ Substituting in Equation (7.82)

$$P = P_1 (1 + D^2) \tag{7.83}$$

or

$$P = P_1 (1 + \text{THD}^2) \tag{7.84}$$

Note:

- The analysis given above can also be performed in terms of the components of output voltage. For distinction, we can use the symbols $A_0, A_1, A_2, A_3, \cdots$ to represent the dc, fundamental, second harmonic, third harmonic etc components respectively of the output voltage.
- The symbols $I_0, I_1, I_2, I_3, I_4, \cdots$ can be equivalently used instead of $B_0, B_1, B_2, B_3, B_4, \cdots$ respectively to represent the components of output current.
- Similarly the symbols $V_0, V_1, V_2, V_3, V_4, \cdots$ can be used instead of $A_0, A_1, A_2, A_3, A_4, \cdots$ respectively to represent the components of output voltage.

Example 7.9

The input excitation of an amplifier is $i_b = I_{bm} \sin \omega t$. Prove that the output current can be represented by a Fourier series which contains only odd sine components and even cosine components.

Solution

From Equation (7.69) the output current is given by

$$i_c = G_1 i_b + G_2 i_b^2 + G_3 i_b^3 + \cdots \tag{A}$$

Given,

$$i_b = I_{bm} \sin \omega t \tag{B}$$

Substituting Equation (B) in Equation (A)

$$i_c = G_1 I_{bm} \sin \omega t + G_2 (I_{bm} \sin \omega t)^2 + G_3 (I_{bm} \sin \omega t)^3 + \cdots$$

$$= G_1 I_{bm} \sin \omega t + G_2 I_{bm}^2 \sin^2 \omega t + G_3 I_{bm}^3 \sin^3 \omega t + \cdots \tag{C}$$

But

$$\sin^2 \omega t = \frac{1 - \cos 2\omega t}{2} \quad \text{and} \quad \sin^3 \omega t = \frac{1}{4}(3 \sin \omega t - \sin 3\omega t) \tag{D}$$

Substituting Equation (D) in Equation (C)

$$i_c = G_1 I_{bm} \sin \omega t + G_2 I_{bm}^2 \frac{1 - \cos 2\omega t}{2} + \frac{G_3 I_{bm}^3}{4} (3 \sin \omega t - \sin 3\omega t) + \cdots$$

$$= G_1 I_{bm} \sin \omega t + \frac{G_2 I_{bm}^2}{2} - \frac{G_2 I_{bm}^2}{2} \cos 2\omega t + \frac{3}{4} G_3 I_{bm}^3 \sin \omega t - \frac{G_3 I_{bm}^3}{4} \sin 3\omega t + \cdots$$

$$= \frac{G_2 I_{bm}^2}{2} + \left(G_1 I_{bm} + \frac{3}{4} G_3 I_{bm}^3 \right) \sin \omega t - \frac{G_3 I_{bm}^3}{4} \sin 3\omega t - \frac{G_2 I_{bm}^2}{2} \cos 2\omega t + \cdots \tag{E}$$

Equation (E) shows that the output current has only odd sine components and even cosine components.

Example 7.10

A single transistor amplifier with transformer coupled load produces harmonic amplitudes in the output as follows:

$$B_0 = 1.5 \text{ mA} \qquad B_3 = 4 \text{ mA}$$
$$B_1 = 120 \text{ mA} \qquad B_4 = 2 \text{ mA}$$
$$B_2 = 10 \text{ mA} \qquad B_5 = 1 \text{ mA}$$

(a) Determine the percentage total harmonic distortion.
(b) Assume that a 2nd identical transistor is used along with a suitable transformer to provide push-pull operation. Use the above harmonic amplitudes to determine the new total harmonic distortion.
(c) If $R_L = 25\ \Omega$, calculate the total power output in each case.

Solution

(a) From Equation (7.77), total harmonic distortion is given by

$$D = \sqrt{D_2^2 + D_3^2 + D_4^2 + D_5^2} \tag{A}$$

$$D_2 = \frac{|B_2|}{|B_1|} = \frac{10}{120} = 0.0833 \text{ or } 8.33\%$$

$$D_3 = \frac{|B_3|}{|B_1|} = \frac{4}{120} = 0.0333 \text{ or } 3.33\%$$

$$D_4 = \frac{|B_4|}{|B_1|} = \frac{2}{120} = 0.0166 \text{ or } 1.66\%$$

$$D_5 = \frac{|B_5|}{|B_1|} = \frac{1}{120} = 0.0083 \text{ or } 0.83\%$$

Substituting in Equation (A),
$$D = \sqrt{0.0833^2 + 0.0333^2 + 0.0166^2 + 0.0083^2}$$
$$= 0.0916 \text{ or } 9.16\%$$

(b) With push-pull connection, *even* harmonic distortion becomes zero (see Section 7.7)

$$\therefore \qquad D_2 = D_4 = 0$$

From Equation (A), the total harmonic distortion is

$$D = \sqrt{D_3^2 + D_5^2}$$
$$= \sqrt{0.0333^2 + 0.0083^2} = 0.0343 \text{ or } 3.43\%$$

Observe that in push-pull configuration, the total harmonic distortion is only 30% of the single ended configuration.

(c) The total power output from Equation (7.83) is

$$P = P_1(1 + D^2) \tag{B}$$

where $P_1 = \dfrac{B_1^2 R_L}{2}$ from Equation (7.79)

$$\therefore \quad P_1 = \frac{(120\times 10^{-3})^2 \times 25}{2} = 0.18 \text{ W}$$

For part (a), $D = 0.0916$

$$\therefore \quad P_1 = 0.18\,(1 + 0.0916^2) = 0.1815 \text{ W}$$

For (b) $D = 0.0343$

$$\therefore \quad P_1 = 0.18\,(1 + 0.0343^2) = 0.1802 \text{ W}$$

Example 7.11

Calculate the harmonic distortion components and the total harmonic distortion for an output signal having fundamental amplitude of 3 V, second harmonic amplitude of 0.3 V, third harmonic amplitude of 0.15 V and fourth harmonic amplitude of 0.06 V. Also find the power delivered by the fundamental component of output voltage if $R_L = 15\Omega$.

Solution

Given $A_1 = 3$ V $A_2 = 0.3$ V $A_3 = 0.15$ V $A_4 = 0.06$ V.

$$D_2 = \frac{|A_2|}{|A_1|} = \frac{0.3}{3} = 0.1 \text{ or } 10\%$$

$$D_3 = \frac{|A_3|}{|A_1|} = \frac{0.15}{3} = 0.05 \text{ or } 5\%$$

$$D_4 = \frac{|A_4|}{|A_1|} = \frac{0.06}{3} = 0.02 \text{ or } 2\%$$

$\therefore$ Total harmonic distortion is

$$D = \sqrt{D_2^2 + D_3^2 + D_4^2}$$

$$= \sqrt{0.1^2 + 0.05^2 + 0.02^2}$$

$$D = 0.1135 \text{ or } 11.35\%$$

Power delivered by the fundamental component of output voltage

$$P_1 = \frac{A_1^2}{2R_L} = \frac{(3\text{V})^2}{(2)(15\Omega)} = 0.3 \text{ W}$$

Example 7.12

The following readings are obtained for a power amplifier.

$$V_{CE\,(\min)} = 2.4\text{ V} \quad V_{CE\,(\max)} = 20\text{ V} \quad V_{CEQ} = 10\text{ V}$$

Calculate the second harmonic distortion.

Solution

From Equation (7.65) using voltages in place of currents, the second harmonic voltage is given by

$$A_2 = \frac{V_{CE(\max)} + V_{CE(\min)} - 2\,V_{CEQ}}{4}$$

$$= \frac{20 + 2.4 - (2\times 10)}{4} = 0.6\text{ V}$$

From Equation (7.64), in terms of voltages, the fundamental output voltage is

$$A_1 = \frac{V_{CE(\max)} - V_{CE(\min)}}{2} = \frac{20 - 2.4}{2} = 8.8\text{ V}$$

$\therefore$ Second harmonic distortion is

$$D_2 = \frac{|A_2|}{|A_1|} = \frac{0.6}{8.8} = 0.0682 \text{ or } 6.82\%$$

Example 7.13

The following readings are obtained for a power amplifier

$$V_{CEQ} = 10\text{ V} \quad V_{\max} = 14\text{ V} \quad V_{\min} = 6\text{ V}$$

Calculate the second harmonic distortion.

Solution

Since, the output is taken between collector and emitter,

$$V_{\max} = V_{CE(\max)} = 14\text{ V}$$

$$V_{\min} = V_{CE(\min)} = 6\text{ V}$$

Fundamental component of output voltage is

$$\therefore \qquad A_1 = \frac{V_{CE(\max)} - V_{CE(\min)}}{2} = \frac{14-6}{2} = 4\text{ V}$$

$$A_2 = \frac{V_{CE(\max)} + V_{CE(\min)} - 2\,V_{CEQ}}{4} = \frac{14+6-20}{4} = 0$$

$\therefore$ Second harmonic distortion is zero.

In this example, since the output voltage swing is symmetric about V_{CEQ}, the amplifier is linear and hence second harmonic distortion is zero.

Example 7.14

For a given power amplifier the output current varies as follows: $I_{max} = 2.5$ A, $I_{min} = 1.8$ A with $I_{CQ} = 2$ A and $R_L = 8\ \Omega$. Find

(a) Second harmonic distortion,
(b) Power delivered by the fundamental component of output current to the load,
(c) Total power delivered to the load.

Solution

$$I_{max} = I_{C\,(max)} = 2.5 \text{ A}$$
$$I_{min} = I_{C\,(min)} = 1.8 \text{ A}$$
$$I_{CQ} = 2 \text{ A} \qquad R_L = 8\ \Omega$$

(a) 2nd harmonic component of current is

$$B_2 = \frac{I_{C(max)} + I_{C(min)} - 2I_{CQ}}{4} = \frac{2.5 + 1.8 - (2 \times 2)}{4} = 0.075 \text{ A}$$

$$B_1 = \frac{I_{C(max)} - I_{C(min)}}{2} = \frac{2.5 - 1.8}{2} = 0.35 \text{ A}$$

∴ Second harmonic distortion

$$D_2 = \frac{|B_2|}{|B_1|} = \frac{0.075}{0.35} = 0.214 \text{ or } 21.4\%$$

(b) Power delivered by the fundamental

$$P_1 = \frac{B_1^2 R_L}{2}$$
$$= \frac{0.35^2 \times 8}{2} = 0.49 \text{ W}$$

(c) Total power delivered to the load

$$P = P_1 + P_2$$
$$P_2 = \frac{B_2^2 R_L}{2} = \frac{0.075^2 \times 8}{2} = 0.0225 \text{ W}$$

Now $\qquad P = 0.49 + 0.0225 = 0.5125$ W.

Example 7.15

For a power amplifier the output current variations are $I_{C(max)} = 5$ A, $I_{C(min)} = 1$ A with $I_{CQ} = 3$ A and $R_L = 5\ \Omega$. Find,

(a) Second harmonic distortion,
(b) Power delivered by the fundamental component of output current and
(c) Total power delivered to the load.

Solution

Output current is the same as collector current

$$\therefore \qquad I_{C(\max)} = 5\text{ A and } I_{C(\min)} = 1\text{ A}$$

(a) Fundamental component of current is

$$B_1 = \frac{I_{C(\max)} - I_{C(\min)}}{2} = \frac{5-1}{2} = 2\text{ A}$$

2nd harmonic component of current is

$$B_2 = \frac{I_{C(\max)} + I_{C(\min)} - 2\,I_{CQ}}{4} = \frac{5+1-(2\times 3)}{4} = 0\text{ A}$$

$$\therefore \qquad D_2 = \frac{|B_2|}{|B_1|} = 0$$

This is expected since the output current is symmetrical and the amplifier is linear.

(b) Power delivered by the fundamental.

$$P_1 = \frac{B_1^2 R_L}{2} = \frac{2^2 \times 5}{2} = 10\text{ W}$$

(c) Power delivered by the second harmonic is 0 W.

$\therefore$ Total power delivered to the load $P = P_1 + P_2 = 10$ W.

Example 7.16

The following distortion readings are available for a power amplifier.

$$D_2 = 0.2 \qquad D_3 = 0.02 \qquad D_4 = 0.06$$

with $\quad I_1 = 3.3$ A $\quad$ and $\quad R_C = 4\ \Omega$

(a) Calculate the THD.
(b) Determine the fundamental power component.
(c) Calculate the total power.

Solution

(a) Total harmonic distortion (THD) is

$$D = \sqrt{D_2^2 + D_3^2 + D_4^2}$$

$$= \sqrt{(0.2)^2 + (0.02)^2 + (0.06)^2} = 0.2097 \text{ or } 20.97\ \%$$

(b) Fundamental component of current is

$$I_1 = B_1 = 3.3\text{ A}$$

$$R_C = R_L = 4\ \Omega$$

$$P_1 = \frac{B_1^2}{2} R_L = \frac{(3.3\,\text{A})^2}{2}(4\ \Omega) = 21.78\ \text{W}$$

(c) $$P = P_1(1 + D^2) = (21.78\ \text{W})\,[1 + (0.2097)^2] = 22.74\ \text{W}$$

7.5 CONVERSION EFFICIENCY

Power amplifiers convert the dc power of the supply V_{CC} into ac (signal) power at the load. The ratio of the ac power delivered to the load to the dc power supplied to the power amplifier is called the **conversion efficiency** or theoritical efficiency. Since, power conversion takes place in the collector circuit of the power transistor, it is also referred to as the collector circuit efficiency. This is a figure of merit for the power amplifier and denoted by η

$$\%\,\eta = \frac{\text{ac or signal power delivered to the load}}{\text{dc power supplied to the amplifier}} \times 100\,\%$$

$$\%\,\eta = \frac{P_{o(\text{ac})}}{P_{i(\text{dc})}} \times 100\,\% \tag{7.85}$$

From Equation (7.10)

$$P_{o(\text{ac})} = \frac{V_{CE(p)}\, I_{C(p)}}{2}$$

The dc power input is given by

$$P_{i\,(dc)} = V_{CC}\, I_{CQ}$$

Substituting in Equation (7.85)

$$\%\,\eta = \frac{V_{CE(p)}\, I_{C(p)}}{2\ V_{CC}\, I_{CQ}} \times 100\,\% \tag{7.86}$$

or $$\%\,\eta = 50\,\frac{V_{CE(p)}\, I_{C(p)}}{V_{CC}\, I_{CQ}}\,\% \tag{7.87}$$

Let us now obtain the conversion efficiency of the series fed and the transformer coupled class A power amplifiers.

7.5.1 Conversion Efficiency of Class A Series-fed Power Amplifier

Refer Fig. 7.8 in Section 7.2.1.

$$V_{CE(p)} = \frac{V_{CE(\max)} - V_{CE(\min)}}{2}$$

Substituting in Equation (7.87)

$$\%\,\eta = \frac{50\left[V_{CE(\max)} - V_{CE(\min)}\right] I_{C(p)}}{2 V_{CC}\, I_{CQ}}\,\%$$

or $$\% \eta = \frac{25\left[V_{CE(\max)} - V_{CE(\min)}\right] I_{C(p)}}{V_{CC}\, I_{CQ}} \%$$

With reference to Fig. 7.8, $I_{C(p)}$ at best can be equal to I_{CQ} for distortionless output as shown in Fig. 7.14.

Substituting $I_{C(p)} = I_{CQ}$ in the above equation.

$$\% \eta = \frac{25\left[V_{CE(\max)} - V_{CE(\min)}\right]}{V_{CC}} \%$$

From the above equation and Fig. 7.14 it is easy to see that efficiency is maximum when

$$V_{CE(\min)} = 0 \text{ and } V_{CE(\max)} = V_{CC}$$

$$\therefore \quad \% \eta_{\max} = 25\% \tag{7.88}$$

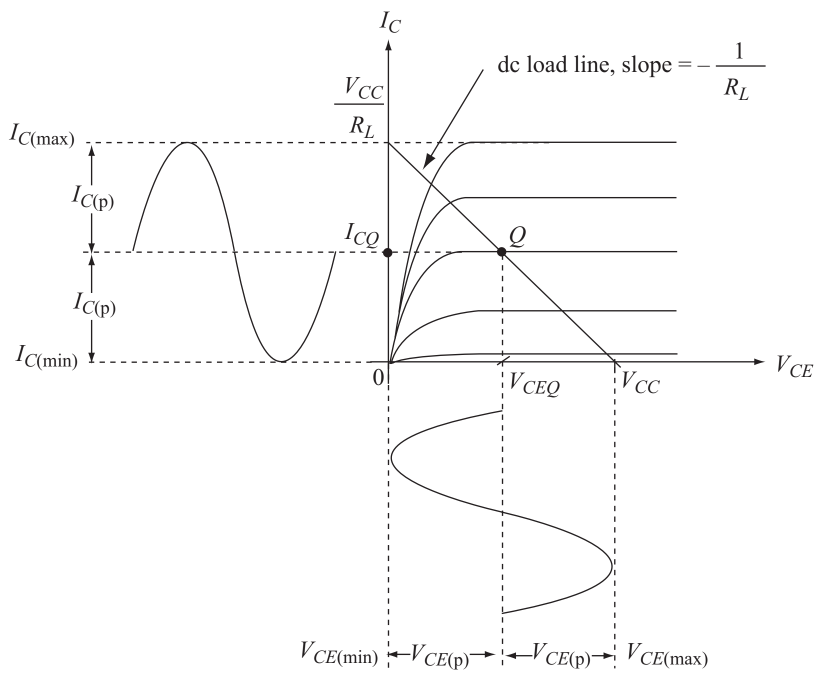


Fig. 7.14 Maximum values of current and voltage for distortionless output

This is the maximum obtainable efficiency for a series-fed class A power amplifier.

The low value of efficiency results from the continuous power dissipation in the collector circuit of the transistor.

7.5.2 Conversion Efficiency of Transformer-Coupled Class A Power Amplifier

Refer Fig. 7.12 in section 7.2.2. From Equations (7.35) and (7.36).

$$V_{CC} = \frac{V_{CE(\max)} + V_{CE(\min)}}{2} \quad \text{and} \quad V_{CE(p)} = \frac{V_{CE(\max)} - V_{CE(\min)}}{2} \tag{7.89}$$

With reference to Fig. 7.12, $I_{C(p)}$ at best can be equal to I_{CQ} for distortionless output.

Substituting $I_{CQ} = I_{C(p)}$ and Equation (7.89) in Equation (7.87)

$$\% \eta = \frac{50 V_{CE(p)} I_{C(p)}}{V_{CC} I_{CQ}} \%$$

$$= 50 \times \frac{\left[\dfrac{V_{CE(\max)} - V_{CE(\min)}}{2}\right]}{\left[\dfrac{V_{CE(\max)} + V_{CE(\min)}}{2}\right]} \%$$

$$\% \eta = 50 \times \frac{V_{CE(\max)} - V_{CE(\min)}}{V_{CE(\max)} + V_{CE(\min)}} \% \tag{7.90}$$

Maximum efficiency can be obtained if $V_{CE(\min)} = 0$

$$\therefore \qquad \% \eta_{\max} = 50 \% \tag{7.91}$$

Thus the maximum attainable efficiency for a transformer-coupled class A power amplifier is 50%. It is double that of the series-fed class A power amplifier.

The higher value of efficiency results from the reduced power dissipation in the collector circuit which is due to the smaller value of dc primary resistance $R_{1\,(dc)}$.

Example 7.17

(a) Calculate the input power, output power and efficiency of the amplifier shown for an input voltage that results in a base current of 5 mA rms. Assume silicon transistor with $\beta = 40$ and $V_{BE} = 0.7$ V.

(b) Show under zero signal condition that the dc power input to the circuit is the sum of dc power dissipated in the load, and power dissipated in the collector.